

# Course 6 – Access Request System (ARS)

## Module 1 – Introduction to Access Request System

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### What is Access Request System (ARS)?

#### Definition

The **Access Request System (ARS)** is a feature in Saviynt that allows users to **request access to applications, roles, accounts, and entitlements** through a centralized and secure process.

Instead of contacting the IT team through emails or phone calls, users can request access directly from Saviynt. The request then follows a predefined approval workflow and, once approved, is provisioned automatically or manually depending on the application type.

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#### Simple Definition

**Access Request System** is a centralized system where users request access, managers approve it, and Saviynt provisions the requested access while maintaining security and audit records.

## 1. Dynamic Approval According to Risk Profiling (Workflow)

#### Definition

A **workflow** is the sequence of approval or rejection steps that an access request follows before access is granted.

Instead of having the same approver every time, **Saviynt decides who should approve based on rules and risk.**

The rules can depend on:

- Application requested
  - Role requested
  - User department
  - Manager
  - Resource owner
  - Account attributes
  - Risk level
  - SoD (Segregation of Duties) violations
- 

### Example 1 (Normal Request)

Employee:

Rahul requests access to Microsoft Teams.

Workflow:

Rahul  
↓  
Manager Approval  
↓  
Access Granted

---

### Example 2 (High Risk Application)

Employee requests SAP Production access.

Workflow:

Employee  
↓  
Manager  
↓  
Application Owner

↓  
Security Team  
↓  
Access Granted

Because SAP Production is high-risk, more approvals are needed.

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## Why is it called Dynamic?

Because approval changes automatically.

Example:

| Request      | Approver                        |
|--------------|---------------------------------|
| Outlook      | Manager                         |
| Salesforce   | Manager + App Owner             |
| Domain Admin | Manager + Security + Compliance |

Saviynt automatically selects the approvers.

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## 2. Pattern Detection

### Definition

Saviynt continuously monitors user behavior.

It checks whether the user is behaving normally or unusually.

It detects:

- Unusual access requests
- Rare activities
- Unexpected login patterns
- Suspicious entitlement usage
- Abnormal behavior

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## Example

Normally:

John  
Logs in:  
9 AM  
From Hyderabad  
Uses Outlook

Today:

2 AM  
From another country  
Requests Admin Access

Saviynt detects this as abnormal.

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Another example:

A user has never requested Database Admin access.

Suddenly they request it.

Saviynt identifies this unusual pattern.

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## Why?

To stop security attacks before damage occurs.

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# 3. Variances in Usage Behavior (Dynamic Risk Scoring)

## Definition

Saviynt gives every identity a **risk score**.

The score changes depending on user behavior.

This is called **Dynamic Risk Scoring**.

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## Static Risk

Assigned once.

Example:

Database Administrator

Risk = High

Always High.

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## Dynamic Risk

Changes over time.

Example:

Yesterday

Risk = 20

Today

Downloaded sensitive data

Logged in from unknown location

Requested privileged role

Risk = 85

The score increases.

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Saviynt also considers:

- Peer group behavior
  - Login history
  - Usage pattern
  - Data sensitivity
  - External security tools
  - Vulnerability reports
- 

## Example

Three HR employees usually access HR Portal.

One employee suddenly accesses Payroll Database.

Saviynt increases the risk score.

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# 4. Accesses are Time-Bound

## Definition

Access is granted only for a limited time.

After that period, access is automatically removed.

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## Why?

To follow the **Least Privilege Principle**.

Least privilege means:

Give users only the access they need and only for as long as they need it.

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## Example

Developer needs Production access.

Instead of permanent access:

Start:  
Monday

End:  
Friday

After Friday:

Saviynt automatically removes access.

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### **Another Example**

An auditor needs access for one week.

After one week,

Access expires automatically.

No manual removal required.

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### **Benefits**

- Better security
  - No unnecessary access
  - Prevents misuse
  - Easier audits
- 

## **5. Persona-Driven Experience**

### **Definition**

A **persona** is a type of user.

Each persona sees only the features required for their job.

Different users have different dashboards and permissions.

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## **Common Personas**

### **1. End User (Requestor)**

Who?

Any employee.

Job:

- Request access
- Check request status
- Reset password

Example:

Rahul wants Salesforce access.

Rahul submits the request.

---

### **2. Manager / Supervisor (Reviewer)**

Who?

Team manager.

Job:

- Approve or reject team members' requests
- Review access

Example:

Rahul requests SAP.

Manager approves it.

---

### **3. Designer / Consultant**



Who?

IAM consultant.

Job:

- Design workflows
- Configure roles
- Build access models

Example:

Creates a workflow:

SAP Access

↓

Manager

↓

Application Owner

↓

Security

---

## **4. Administrator**

Who?

Saviynt Admin.

Job:

- Configure platform
  - Install connectors
  - Schedule jobs
  - Import users
  - Troubleshoot issues
- 

## **5. Application Owner**

Who?

Person responsible for an application.

Job:

- Onboard applications
- Manage application lifecycle
- Review application access

Example:

Owns Salesforce.

Approves Salesforce requests.

---

## **6. Resource Owner**

Who?

Owner of accounts, roles, and entitlements.

Job:

- Manage permissions
- Decide who gets access

Example:

Owns Finance Roles.

Approves Finance Role requests.

---

## **7. Security & Compliance Owner**

Who?

Security team.

Job:

- Create security policies
- Configure SoD rules
- Define risk policies

Example:

Creates rule:

No user can have:

Accounts Payable

+

Accounts Receivable

---

## **8. Auditor**

Who?

Audit team.

Job:

Checks whether:

- Policies are followed
- Users have correct access
- Violations exist

Example:

Verifies that terminated employees no longer have access.

---

## **9. Help Desk**

Who?

Support team.

Job:

- Password resets
- Unlock accounts
- Basic troubleshooting

Example:

Employee cannot log in.

Help Desk unlocks the account.

---

## 6. Approval Details Page (Expanded View)

### Definition

When an approver opens an access request, the **Approval Details** page is expanded by default.

This means the approver immediately sees all important information without clicking multiple sections.

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### The approver can easily see:

- Who requested access
  - Which application
  - Which role or entitlement
  - Business justification
  - Risk score
  - Manager
  - Resource owner
  - Approval history
  - SoD violations (if any)
- 

### Example

Rahul requests SAP Admin.

The approver opens the request and immediately sees:

Requester:

Rahul

Application:

SAP

Role:  
SAP Admin

Reason:  
Project Work

Risk:  
High

Manager:  
Approved

Application Owner:  
Pending

This helps the approver make a quick and informed decision.

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## Quick Revision Table

| Feature                          | Meaning                                                            | Example                                                                       |
|----------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Dynamic Workflow</b>          | Approval path changes based on rules and risk.                     | Outlook → Manager only; SAP Admin → Manager + Security + App Owner            |
| <b>Pattern Detection</b>         | Detects unusual user behavior.                                     | User logs in at 2 AM from another country and requests admin access.          |
| <b>Dynamic Risk Scoring</b>      | Risk score changes based on user activity.                         | Suspicious downloads increase the user's risk score.                          |
| <b>Time-Bound Access</b>         | Access expires automatically after a defined period.               | Temporary production access removed after one week.                           |
| <b>Persona-Driven Experience</b> | Each user sees only the features needed for their role.            | End User requests access, Manager approves, Admin configures the platform.    |
| <b>Approval Details Page</b>     | Shows complete request information in an expanded view by default. | Approver immediately sees requester, application, risk, and approval history. |

## **Interview Question**

**Q: Why does Saviynt use dynamic workflows instead of fixed approval workflows?**

**Answer:** Dynamic workflows automatically choose the appropriate approvers based on factors such as the application, requested role, user attributes, resource owner, risk level, and SoD policies. This ensures that high-risk requests receive additional scrutiny while low-risk requests are processed more efficiently.

# Access Request

Think of an **Access Request** in Saviynt like **ordering something online**.

- You place the order (**Request**).
- Someone approves it (**Manager/Owner**).
- The item is delivered (**Access Provisioned**).

The complete process has **3 major stages**.

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## Overall Flow

User Needs Access



1. User Submits Request



2. Approver Reviews Request



3. Saviynt Fulfills Request



User Gets Access

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## Step 1: User Submits the Access Request

### Goal:

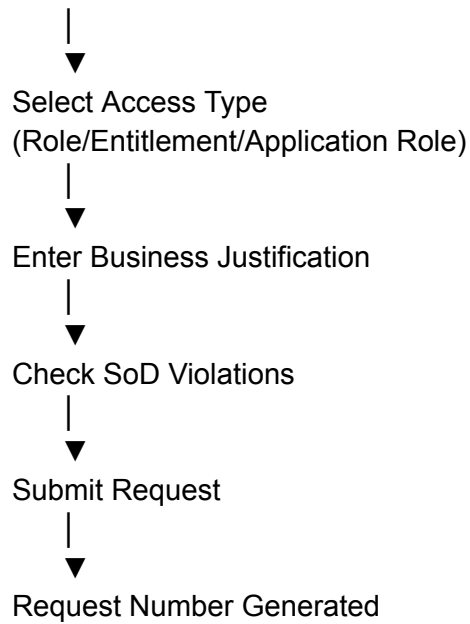
The user tells Saviynt what access they need.

### Process

Login to Saviynt



Select Application



## Explanation of each step

### 1. Login

The user logs into Saviynt.

Example

Employee: Rahul

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### 2. Select the Application

Choose the application for which access is needed.

Example

SAP  
Salesforce  
Active Directory  
ServiceNow

---



### 3. Select Access Type

The user selects what exactly they need.

Possible options:

- Role
- Entitlement
- Application Role
- Enterprise Role

Example

Application:  
SAP

Access:  
Finance Role

---

### 4. Provide Business Justification

The user explains **why** they need access.

Example

Need SAP Finance role  
to work on monthly reports.

Without justification, approvers may reject the request.

---

### 5. SoD Check

Before submission, Saviynt checks whether granting this access will violate **Segregation of Duties (SoD)**.

Example

Current Role

Payment Approver

Requested Role

Payment Creator

Having both roles is an SoD violation.

Saviynt displays a warning.

The user may add **mitigating controls** if required.

---

## 6. Submit

After clicking **Submit**:

Saviynt

- Creates a Request ID
- Sends email notification
- Shows request number

Example

Request Number

REQ00012345

---

## Step 2: Approver Reviews the Request

Now the request moves to the approver.

Request Submitted



Manager/Application Owner  
receives notification



Reviews request





Takes Action

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## How does the approver know?

Saviynt sends notifications through:

- Web UI
  - Mobile App
  - Email
- 

## What does the approver review?

The approver checks:

- Who requested?
- Which application?
- Which role?
- Business justification
- Risk score
- SoD violations

Example

Employee:  
Rahul

Application:  
SAP

Role:  
Finance

Reason:  
Project Work

Risk:  
Medium

SoD:

None

---

# What actions can the approver take?

## 1. Approve

Means:

"Yes, give the requested access."

Flow

Approve



Provisioning Starts

---

## 2. Reject

Means

"No, don't provide access."

Example

Business justification is insufficient.

---

## 3. Modify

The approver changes the request.

Example

Instead of

SAP Admin

approve only

SAP Read Only

What happens?

Old Request

↓

Cancelled

↓

New Modified Request Created

---

## 4. Reroute

Approver forwards the request to someone else.

Example

Manager

↓

Application Owner

---

## 5. Consult

The approver asks an expert for advice before deciding.

Example

Manager

↓

Security Team



Manager Approves

---

## 6. Discontinue

Stops the request completely.

Used when

- Wrong request
- Duplicate request
- User no longer needs access

Only possible while the request is still **Open**.

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## After the Decision

Saviynt immediately notifies the user.

Example

Your request has been

Approved

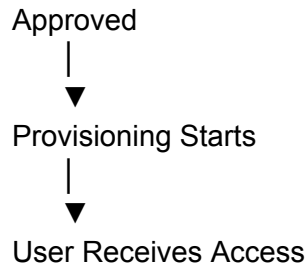
or

Rejected

---

## Step 3: Request Fulfillment (Provisioning)

Now Saviynt actually gives access.



This is called **Provisioning**.

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## What happens internally?

Saviynt creates a **Task**.

Example

Create Account

Assign Role

Add Entitlement

Saviynt tracks whether the task is completed.

---

## Two Types of Applications

This is the most important interview question.

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## Connected Application

### Meaning

Saviynt is directly connected to the application using a **connector**.

Everything happens automatically.

Flow

Request



Approval



Connector



Application



Access Granted Automatically

Example

Active Directory

Azure

Salesforce

Office 365

---

## Real Example

Rahul requests Active Directory access.

Submit



Manager Approves



↓

Connector Runs

↓

AD Account Created

↓

Rahul Gets Access

No human intervention.

---

## Disconnected Application

### Meaning

Saviynt has **no direct connector**.

Humans must perform the access manually.

Flow

Request

↓

Approval

↓

Saviynt Creates Task

↓

Ticket Created

↓

Admin Gives Access Manually



Update Sent Back to Saviynt



Task Closed

---

## Example

Application

Legacy Payroll System

No connector exists.

After approval:

Saviynt



Creates Task



Creates ServiceNow Ticket



IT Team Opens Ticket



Creates User Manually



Marks Ticket Completed



Saviynt Closes Task

---

## ServiceNow (SNOW) Flow

For disconnected applications:

Approval



Saviynt Task Created



ServiceNow Ticket Created



Provisioner Works on Ticket



Access Given



Status Updated to Saviynt



Task Closed

---

## WSRETRYJOB (Connected Applications)

For connected applications, Saviynt uses a background job called **WSRETRYJOB**.

Its job is to:

- Provision access
- Remove access (deprovision)
- Retry failed provisioning tasks
- Mark tasks as completed after successful execution

Example

Approval



WSRETRYJOB Runs



Creates AD Account



Marks Task Completed

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## Complete End-to-End Flow

Employee Needs Access



Login to Saviynt



Select Application

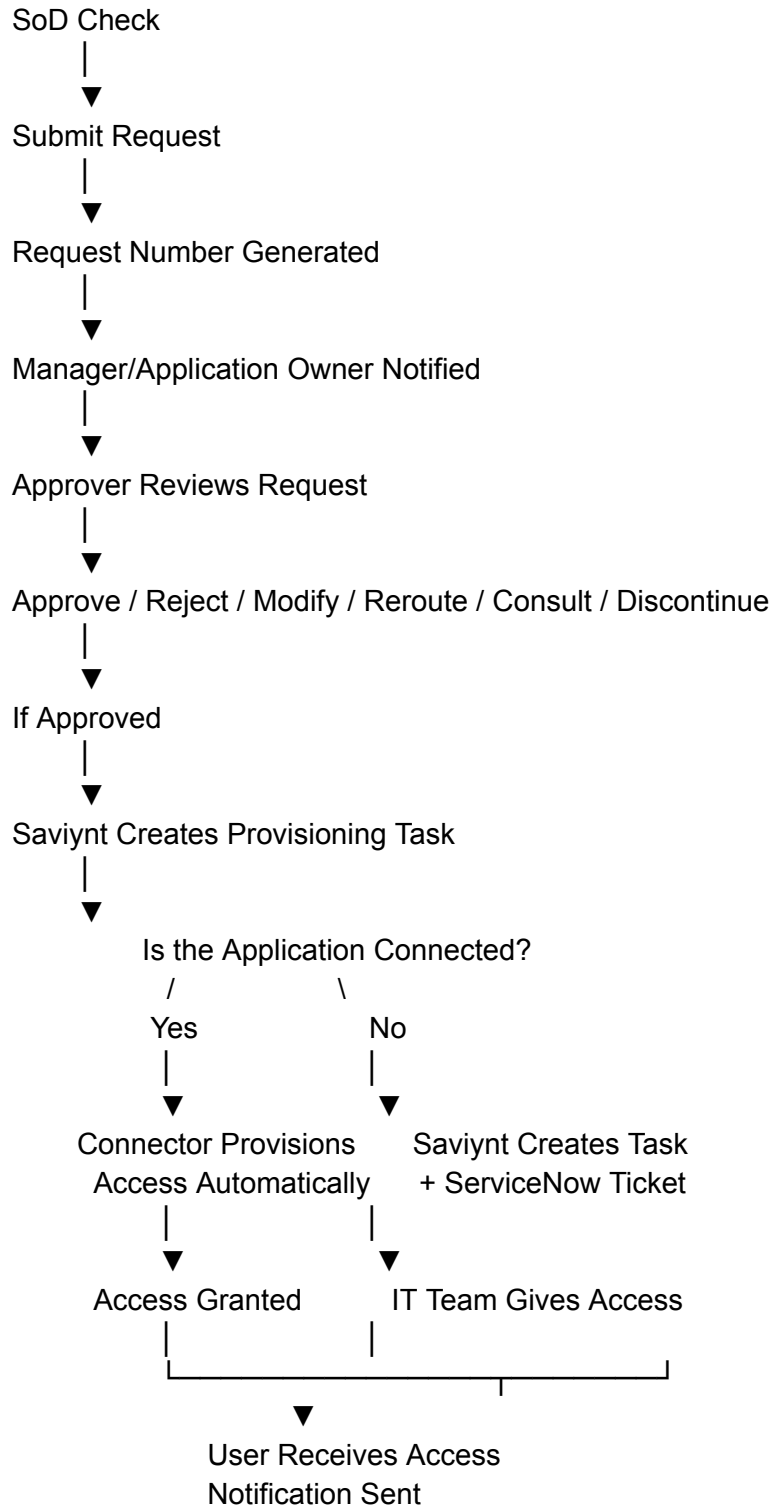


Select Role/Entitlement



Provide Business Justification





**Interview Summary (Remember These 3 Steps)**

1. **Request** – The user logs into Saviynt, selects the application and required access, provides a business justification, Saviynt checks for SoD violations, and the request is submitted.
2. **Approval** – The manager, application owner, or another approver reviews the request and can **Approve, Reject, Modify, Reroute, Consult, or Discontinue** it.
3. **Fulfillment (Provisioning)** – If approved:
  - **Connected application:** Saviynt provisions the access automatically through a connector (using jobs like **WSRETRYJOB**).
  - **Disconnected application:** Saviynt creates a task (and often a **ServiceNow** ticket), an administrator grants access manually, and the task is closed after completion.

## Quick Revision

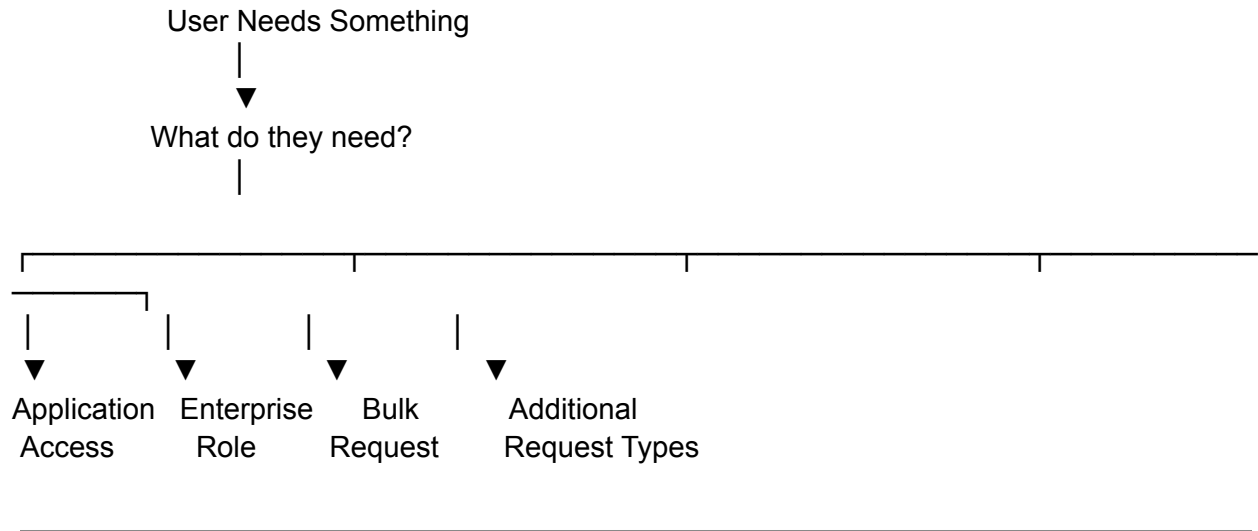
- **New Access Request** → Grants new access.
- **Modify Access Request** → Changes existing permissions.
- **Remove Access Request** → Revokes access.
- **Account Request** → Creates an application account.
- **Entitlement Request** → Assigns a specific permission or group.
- **Role Request** → Assigns multiple permissions through a role.
- **Application Request** → Grants access to an application.
- **Birthright Access** → Automatically assigned based on business rules.
- **Temporary Access** → Time-limited access that expires automatically.
- **Emergency (JIT) Access** → Temporary privileged access for urgent situations.
- **Password Management Request** → Password reset, password change, or account unlock.
- Access requests can also be **Self, Request for Others, Bulk, or Delegated** depending on who initiates the request.

## Best Practices

- Use Birthright Access for mandatory applications.
- Use Role Requests instead of individual entitlement requests where possible.
- Grant Temporary Access instead of permanent privileged access.

- Use Emergency Access only for production incidents.
- Regularly review requestable applications and roles.
- Configure approval workflows based on risk.
- Enable SoD validation before provisioning.
- Maintain audit logs for every request

# Types of Access Requests in Saviynt



## 1. Application Access Request (Most Common)

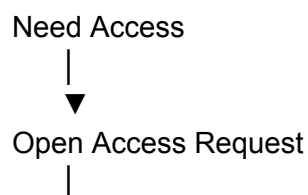
### Definition

An **Application Access Request** is used when a user needs access to a **specific application** or something inside that application.

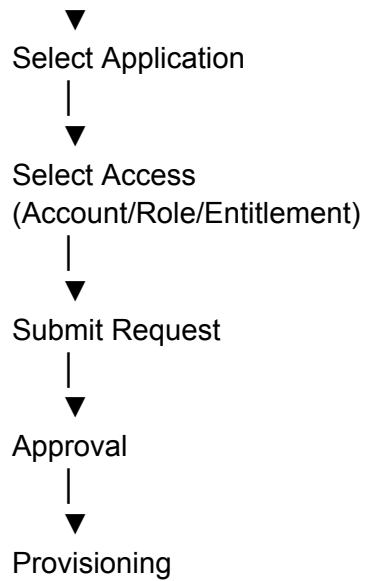
The user can request:

- User Account
- Entitlement
- Privilege
- Application Role

### Flow



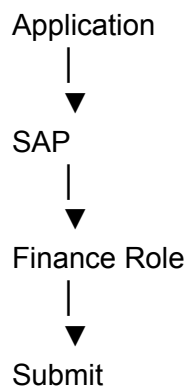




## Example 1

Employee Rahul joins the Finance team.

He needs SAP access.



## Example 2

Employee needs Salesforce access.

Application

Salesforce



Sales User Role



Submit Request

---

## What can be requested?

| Request          | Example           |
|------------------|-------------------|
| Account          | Create AD Account |
| Entitlement      | Read Reports      |
| Privilege        | Printer Access    |
| Application Role | SAP Finance Role  |

---

## Remember

Application Access = **One Application**

---

## 2. Enterprise Role Access Request

### Definition

An **Enterprise Role** is a bundle of access across **multiple applications**.

Instead of requesting each application separately, request one Enterprise Role.

---

## Flow

Need Multiple Accesses



Select Enterprise Role



Submit



Approval



All Required Accesses Granted

---

## Example

New HR employee joins.

Instead of requesting:

- Workday
- Outlook
- Teams
- Active Directory

one by one,

Saviynt has

Enterprise Role

HR Employee

This role automatically contains

HR Employee Role



— Workday

— Outlook

└ Teams  
└ Active Directory

User requests only:

HR Employee Role

Saviynt grants everything.

---

## Why use Enterprise Roles?

Instead of

SAP

+

AD

+

Teams

+

VPN

Request only

Finance Employee Role

Everything is included.

---

## Remember

Enterprise Role = **Multiple Applications bundled together**

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## 3. Bulk Request

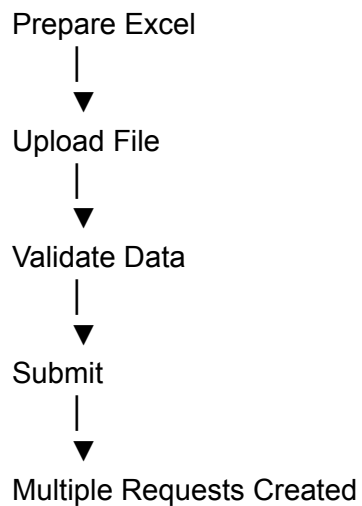
### Definition

A **Bulk Request** allows requesting access for **multiple users at once**.

Instead of creating requests one by one, upload an Excel file.

---

### Flow



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### Example

HR hires

50 Employees

Instead of

Employee 1

↓

Employee 2



Employee 3



...



Employee 50

Use one Excel file.

Excel File

Rahul

Priya

Amit

John

David

Upload once.

Saviynt creates all requests.

---

## Benefits

- Saves time
  - No repetitive work
  - Easy for onboarding many employees
-

## Remember

Bulk Request = **Many Users**

---

## Comparison of the Three Main Types

| Type               | Used For                       | Example               |
|--------------------|--------------------------------|-----------------------|
| Application Access | One application                | SAP Access            |
| Enterprise Role    | Multiple applications together | HR Employee Role      |
| Bulk Request       | Multiple users                 | Onboard 100 employees |

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## Additional Access Request Types

These are special request types.

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## 4. Emergency Role Request

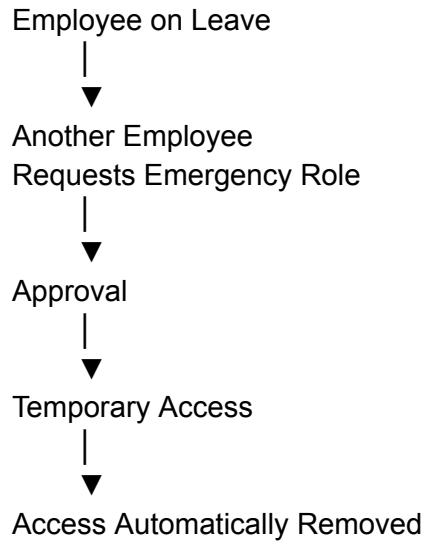
### Definition

Temporary role assignment for someone covering another employee.

Usually used when someone is:

- On leave
  - Sick
  - Vacation
- 

### Flow



## Example

Manager is on leave.

Rahul needs to approve invoices.

Rahul requests

Emergency Finance Manager Role

Valid for

7 Days

After 7 days

Access is removed automatically.

---

## Remember

Emergency Role = **Temporary Job Role**

---



# 5. Emergency ID Request

## Definition

Request a **Privileged Account (Admin Account)** for emergencies.

Usually for:

- Server failure
- Production issue
- Critical incident

---

## Flow

Emergency Occurs



Request Privileged ID



Approval



Temporary Admin Access



Access Removed

---

## Example

Production server crashes.

Engineer requests

Windows Admin Account

Gets access for

2 Hours

After 2 hours

Access expires automatically.

---

## Remember

Emergency ID = **Temporary Admin Account**

---

# 6. Group Management Request

## Definition

Used to create or modify groups.

Example

Create

Active Directory Group

HR\_Managers

or

Add Users

to Existing Group

---

## 7. Service Account Request

### Definition

Used to request a **Service Account**.

A service account is **not used by humans**.

It is used by:

- Applications
  - Scripts
  - Services
  - Scheduled Jobs
- 

### Example

Payroll application needs to connect to SQL Server.

It uses

Payroll\_Service\_Account

instead of a human user.

---

## 8. Role Management Request

### Definition

Used to modify existing roles.

Example

Current Role

Finance Role

Need to add

Invoice Approval

Submit

Role Modification Request

---

## 9. User Management Request

### Definition

Used to manage user identities.

Examples

- Create User
  - Modify User
  - Disable User
  - Delete User
- 

### Example

New employee joins.

Request

Create User

Employee leaves.

Request

Disable User

---

# 10. Organization Management Request

## Definition

Used to manage organizational information.

Examples

- Create Department
  - Modify Department
  - Add Business Unit
  - Update Organization Structure
- 

## Easy Trick to Remember

| Request Type                   | Think Of It As                                                  |
|--------------------------------|-----------------------------------------------------------------|
| <b>Application Access</b>      | One application for one user                                    |
| <b>Enterprise Role</b>         | A package of access across multiple applications                |
| <b>Bulk Request</b>            | Same request for many users using Excel                         |
| <b>Emergency Role</b>          | Temporary role replacement when someone is absent               |
| <b>Emergency ID</b>            | Temporary privileged/admin account for emergencies              |
| <b>Group Management</b>        | Create or modify user groups                                    |
| <b>Service Account</b>         | Non-human account used by applications or services              |
| <b>Role Management</b>         | Create or modify roles                                          |
| <b>User Management</b>         | Create, update, disable, or delete user accounts                |
| <b>Organization Management</b> | Manage departments, business units, or organizational structure |

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# Interview Question

**Q: What is the difference between an Application Access Request and an Enterprise Role Request?**

## **Application Access Request**

Grants access to **one application**.

Example: Request **SAP Finance Role** only.

## **Enterprise Role Request**

Grants access to **multiple applications** bundled into one business role.

Example: Request **Finance Employee Enterprise Role**, which automatically includes SAP, Active Directory, Outlook, Teams, and other required access.

**In one sentence:**

- **Application Access** = Access to a **single application**.
- **Enterprise Role** = A **bundle of access** across multiple applications requested with a single request.

# Workflow in Saviynt (Simple Explanation)

Think of a **workflow** as a **road map** that tells Saviynt:

**"After a user submits an access request, who should approve it, in what order, and under what conditions?"**

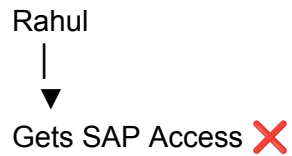
It is simply the **approval path** that a request follows until it is approved or rejected.

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## Simple Real-Life Example

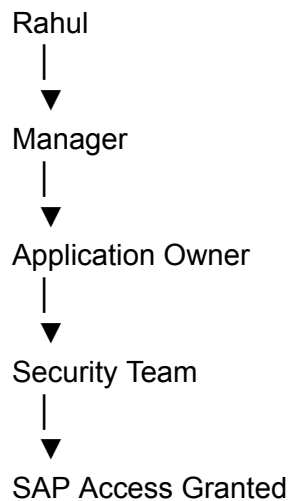
Suppose Rahul wants SAP access.

Without a workflow:



Anyone could get access, which is insecure.

With a workflow:



The workflow ensures that the right people approve the request before access is given.

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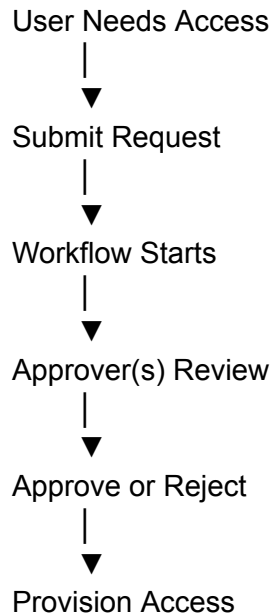
## Definition

A **workflow** is a sequence of approval or rejection steps that an access request follows after it is submitted.

Its purpose is to ensure that the correct people review and approve the request before access is granted.

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# Overall Workflow Process



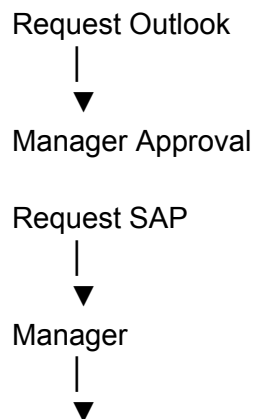
## Workflow Can Be Based On

Saviynt decides the approval path based on different conditions.

### 1. Application-Based Workflow

Approval depends on the application being requested.

#### Example





SAP Owner

Different applications can have different workflows.

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## 2. Role-Based Workflow

Approval depends on the requested role.

### Example

Request Employee Role



Manager

Request Admin Role



Manager



Security Team

Higher-privilege roles require more approvals.

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## 3. Account Attribute-Based Workflow

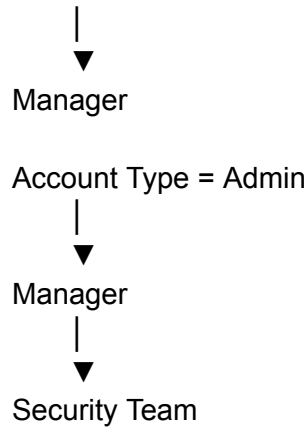
Approval depends on account attributes.

Account attributes are properties of the account, such as:

- Account Type
- Privilege Level
- Environment (Production/Test)
- Account Status

### Example

Account Type = Normal



## 4. User Attribute-Based Workflow

Approval depends on user information.

User attributes include:

- Department
- Location
- Designation
- Employee Type

### Example

Department = HR



Department = Finance



Different users follow different approval paths.

---

## Dynamic Workflow

# What is Dynamic?

The approval process changes automatically depending on the request.

Saviynt evaluates the request and selects the appropriate workflow.

---

## Example 1 (Low Risk)

Rahul requests Microsoft Teams.

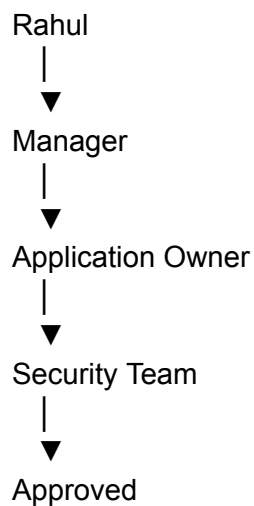


Only one approval.

---

## Example 2 (High Risk)

Rahul requests SAP Admin.



Because the request is high risk, additional approvals are required.

---

# Workflow Supports SoD

SoD (Segregation of Duties) prevents users from receiving conflicting access.

## Example

Current Role:

Payment Creator

Requested Role:

Payment Approver

A user should not have both roles.

Saviynt detects this SoD violation and can send the request for additional approval or reject it.

---

# Types of Workflows

Saviynt supports two workflow types.

---

## 1. Serial Workflow

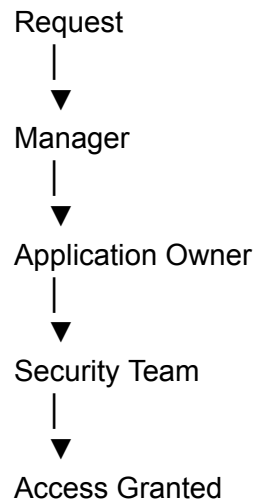
### Definition

Approvals happen **one after another**.

The next approver receives the request only after the previous approver has approved it.

---

## Flow

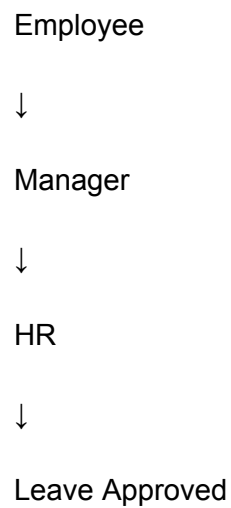


If the Manager rejects the request, it never reaches the Application Owner.

---

## Real-Life Example

Applying for leave:



Each step waits for the previous one.

---

## 2. Parallel Workflow

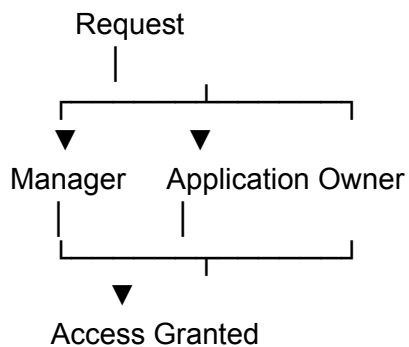
### Definition

Multiple approvers receive the request at the same time.

Everyone can review it simultaneously.

---

### Flow



Both approvals happen in parallel.

---

### Example

A Production Database request requires approval from:

- Manager
- Database Owner

Both receive the request at the same time, reducing waiting time.

---

## Serial vs Parallel Workflow

### Serial Workflow

One approval after another

### Parallel Workflow

Multiple approvals simultaneously

Slower

Next step starts only after the previous one finishes

Example: Manager → Security

Faster

All approvers can review at the same time

Example: Manager and Security together

---

# Workflow for SOX Compliance

## What is SOX?

**SOX (Sarbanes-Oxley Act)** is a law that helps organizations protect financial systems by enforcing proper controls and approvals.

In Saviynt, SOX workflows ensure that sensitive financial access receives additional review.

---

## SOXCRITICALCOUNT

Saviynt uses the variable:

SOXCRITICALCOUNT

It stores the number of SOX-critical violations detected in the request.

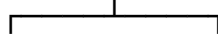
---

### Logic

Request Submitted



Check SOXCRITICALCOUNT



= 0



> 0



▼      ▼  
Manager   SOX Compliance Team  
Approval   Approval

---

### **Example 1 (No SOX Violation)**

SOXCRITICALCOUNT = 0

Flow:

Employee

↓

Manager

↓

Access Granted

---

### **Example 2 (SOX Violation)**

SOXCRITICALCOUNT = 2

Flow:

Employee

↓

SOX Compliance Team

↓

Manager

↓

Access Granted



Additional approval is required because the request is considered sensitive.

---

# Approval Logic Process

Every workflow follows three main stages.

---

## Step 1: Create the Workflow

An administrator designs the workflow.

They define:

- Who should approve
- Approval sequence
- Conditions
- Rules

Workflow creation can be done using:

- Graphical Workflow Editor (drag-and-drop)
  - XML Editor
- 

## Example

Create Workflow

↓

If SAP

↓

Manager

↓

Application Owner



Security

---

## Step 2: Approve the Request

When a user submits a request, the workflow starts.

Approvers can:

- Approve
  - Reject
  - Modify
  - Reroute
  - Search for requests
- 

### Example

Manager



Approves



Application Owner Receives Request

---

## Step 3: Assign the Workflow

The created workflow is attached to an application, role, endpoint, or other resource.

From that point onward, every request for that resource follows the assigned workflow.

---

### Example

Workflow Created



Assigned to SAP



Every SAP Request



Follows Same Approval Process

---

## Complete Workflow Process

Admin Creates Workflow



Workflow Assigned to Application/Role



User Requests Access



Workflow Starts



Evaluate Conditions  
(Application, Role, User Attributes,  
Account Attributes, Risk, SoD)



Select Correct Approval Path

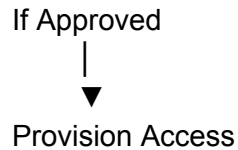


Serial or Parallel Approvals



Approve or Reject





## Quick Interview Revision

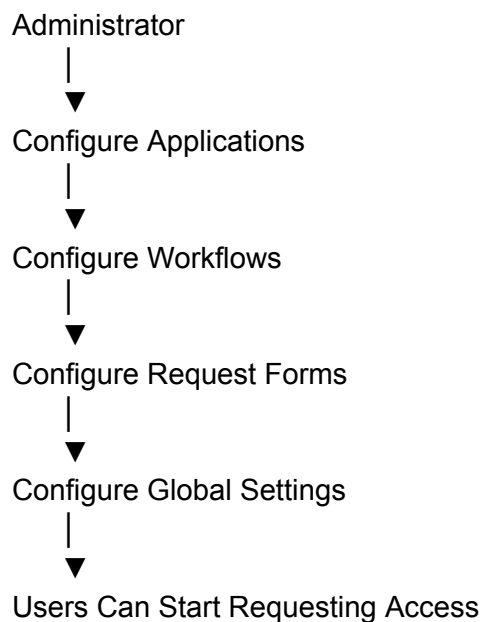
| Concept                                 | Meaning                                                                                                                               | Example                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Workflow</b>                         | The approval path followed by an access request.                                                                                      | User → Manager → Application Owner                                                                        |
| <b>Dynamic Workflow</b>                 | Approval path changes automatically based on conditions such as application, role, user attributes, account attributes, risk, or SoD. | SAP Admin requires Manager + Security, while Outlook requires only Manager approval.                      |
| <b>Application-Based Workflow</b>       | Approval depends on the requested application.                                                                                        | SAP requests go to the SAP Owner.                                                                         |
| <b>Role-Based Workflow</b>              | Approval depends on the requested role.                                                                                               | Admin roles require Security approval.                                                                    |
| <b>User Attribute-Based Workflow</b>    | Approval depends on user information such as department or location.                                                                  | Finance users go to the Finance Manager.                                                                  |
| <b>Account Attribute-Based Workflow</b> | Approval depends on account properties such as account type or privilege level.                                                       | Admin accounts require extra approval.                                                                    |
| <b>Serial Workflow</b>                  | Approvals happen one after another.                                                                                                   | Manager → Security → Application Owner.                                                                   |
| <b>Parallel Workflow</b>                | Multiple approvers review simultaneously.                                                                                             | Manager and Application Owner approve at the same time.                                                   |
| <b>SOX Workflow</b>                     | Uses <b>SOXCRITICALCOUNT</b> to decide whether additional SOX compliance approval is needed.                                          | If <b>SOXCRITICALCOUNT &gt; 0</b> , the request is routed to the SOX Compliance team before provisioning. |

This topic explains **how an administrator configures Access Requests in Saviynt**. Think of it as **setting the rules before users start requesting access**.

---

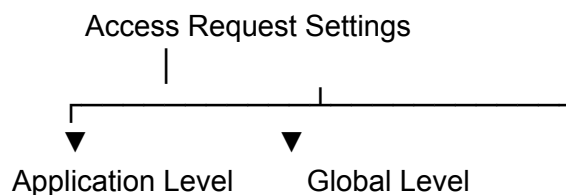
## Setting Up Access Requests (Overall Flow)

Before users can request access, the administrator configures the system.



## Two Types of Settings

Saviynt has two main configuration levels.



# 1. Application-Level Settings

These settings apply to **one specific application**.

Example:

Suppose your company has:

- SAP
- Salesforce
- Active Directory

Each application can have its own settings.

---

## A. Security System (Workflow)

### Purpose

Define **who should approve requests** for that application.

Example

For Outlook

Request Outlook



Manager Approval

For SAP

Request SAP



Manager



SAP Owner



Security Team

Different applications can have different approval workflows.

---

## B. Endpoint (Request Form)

### Purpose

Customize what the user sees while requesting access.

The administrator decides:

- Which fields appear
- Which fields are mandatory
- Which access options are displayed

Example

SAP Request Form

Application:

SAP

Department:

Finance

Reason:

---

Project Name:

---

Salesforce may have a different request form.

---

## 2. Global Settings

These settings apply to **the entire Saviynt system**, not just one application.

Examples:

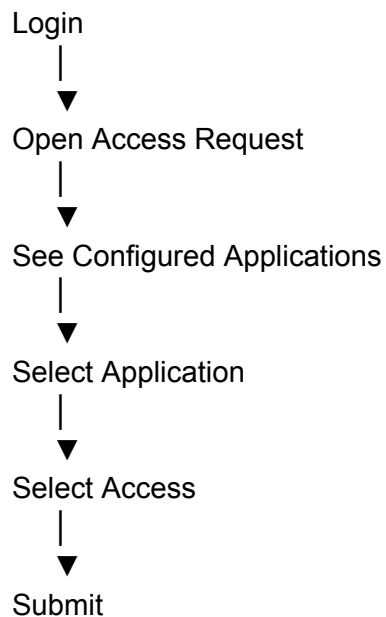
- Comments mandatory
- Start date
- End date
- Account name entry
- Request limits

Every application follows these settings.

---

## Application Access Request Process

This is what the user experiences.



Example

Rahul

↓

SAP

↓

Finance Role



↓

Submit

---

# Features of Application Access Requests

These are the features available when requesting access.

---

## 1. Risk-Based Dynamic Approval Flow

### Meaning

Approval changes based on the request's risk.

Example

Low Risk

Teams

↓

Manager

High Risk

SAP Admin

↓

Manager

↓

Security



Application Owner

More risk means more approvals.

---

## 2. Intelligent Recommendations (JRM)

### Meaning

Saviynt recommends access based on what similar employees already have.

This is called **Peer Access Analysis**.

---

### Example

Five HR employees already have:

Workday

Outlook

Teams

A new HR employee joins.

Saviynt recommends:

Recommended Access

✓ Workday

✓ Outlook

✓ Teams

Instead of selecting manually.

---

## 3. Security Policies

### Meaning

Users must agree to company security policies before submitting requests.

Example

☐

I agree to use this access only for business purposes.

Only after checking the box can the request be submitted.

---

## 4. Preventive SoD Check

### Meaning

Saviynt checks for **Segregation of Duties (SoD)** conflicts before the request is submitted.

---

Example

Current Role

Invoice Creator

Requested Role

Invoice Approver

Saviynt displays:

⚠ SoD Violation Detected

The user can review the warning before submitting.

---

## 5. Simplified User Interface

### Meaning

Saviynt provides a simple and easy interface.

Users can:

- Search applications
- Select access
- Submit requests

without needing technical knowledge.

---

## 6. Easy-to-Set-Up Forms

### Meaning

Administrators create custom request forms.

Different applications can ask for different information.

---

Example

VPN Request

Reason

Manager

Duration

SAP Request

Department

Project

Cost Center

These are created using **Dynamic Attributes**.

---

## What are Dynamic Attributes?

Dynamic attributes are **additional fields** added to a request form.

Example

Instead of only asking:

Application

the administrator adds:

Project Name

Cost Center

Business Unit

These extra fields are called **Dynamic Attributes**.

---

## 7. Approval Confidence

### Meaning

Saviynt compares the requested access with peer access groups.

If many similar users already have that access,

Saviynt gives the approver greater confidence to approve.

---

Example

All Finance employees have:

SAP Finance

Excel

Power BI

A new Finance employee requests SAP Finance.

Saviynt indicates that this is a common request.

The approver is more confident approving it.

---

## Filter Existing Access

### Problem

Users may already have some applications.

Without filtering:

SAP

Salesforce

Outlook

Teams

VPN

The user doesn't know which ones they already have.

---

## **Solution**

Saviynt hides applications the user already has.

Now only missing applications appear.

VPN

Power BI

This reduces confusion.

---

## **Allow Adding Account Name**

Configuration Path

Admin

↓

Global Configuration

↓

Request

↓

Allow Adding Account Name From Request Form

---

## **Purpose**

Allows users to type the account name during the request.

Example

Instead of automatic account creation,

the user enters:

Account Name

rahul123

---

## Management of Inactive Accounts

Sometimes a user account already exists but is disabled.

Example

Rahul\_AD

Status

Inactive

Previously

The user couldn't request changes easily.

Now

Saviynt allows:

- Modify the inactive account
  - Keep it inactive
  - Enable it if needed
- 

## Show Inactive Accounts

When enabled,

Saviynt displays inactive accounts during the request.

Example

Rahul\_AD



Inactive

Now the requester can decide what to do.

---

## Enable Inactive Account During Request

Flow

Inactive Account



Request Access



Choose

Enable Account?

Yes / No

---

Example

Current Status

Account Disabled

Administrator chooses

Enable Account

+

Assign SAP Access

Both happen together.

---

# Modify Disabled Accounts

Example

Disabled Account



Assign New Role



Enable Account



Complete Request

Or

Disabled Account



Keep Disabled



Only Add Future Access

Saviynt supports both options.

---

## Request Guardrails

Think of **Guardrails** as **limits or restrictions**.

They prevent users from making unreasonable requests.

Configuration Path

Admin



Global Configuration



Request



Advanced Settings



Request Guardrails

---

# Guardrail 1

Maximum Applications Per Request

Default

10 Applications

Example

The user selects:

SAP

AD

VPN

Salesforce

Teams

Power BI

Allowed.

But selecting

25 Applications

is blocked.

---

## Guardrail 2

Maximum Enterprise Roles

Default

20 Enterprise Roles

---

## Guardrail 3

Maximum Access Items

Default

50 Items

Example

A user requests:

- Roles
- Entitlements
- Privileges

Total:

48

Allowed

Total:

60

Rejected

---

## Guardrail 4

Maximum Comment Length

Default

1000 Characters

Users cannot enter comments longer than the configured limit.

---

## Complete Process Flow

Administrator



Configure Workflow



Configure Request Form

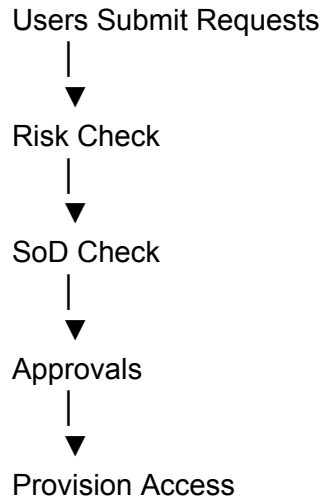


Configure Global Settings



Configure Guardrails





## Quick Revision Table

| Feature                                  | Purpose                                                 | Example                                                  |
|------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| <b>Application-Level Settings</b>        | Configure settings for a specific application.          | SAP has its own workflow and request form.               |
| <b>Global Settings</b>                   | Configure settings that apply to all requests.          | Make comments mandatory or require start/end dates.      |
| <b>Workflow</b>                          | Defines who approves requests.                          | Manager → Application Owner → Security.                  |
| <b>Endpoint Request Form</b>             | Customizes the fields shown to users.                   | Ask for Project Name and Cost Center for SAP requests.   |
| <b>Risk-Based Dynamic Approval</b>       | Changes approval path based on request risk.            | High-risk requests require additional approvers.         |
| <b>Intelligent Recommendations (JRM)</b> | Suggests access based on similar users (peer analysis). | Recommend Workday and Teams to a new HR employee.        |
| <b>Preventive SoD Check</b>              | Detects conflicting access before submission.           | Warn if a user requests both Creator and Approver roles. |
| <b>Dynamic Attributes</b>                | Extra fields added to request forms.                    | Project Name, Business Unit, Cost Center.                |

|                                    |                                                                 |                                                                                  |
|------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Inactive Account Management</b> | Allows modifying or enabling inactive accounts during requests. | Enable a disabled account and assign new access in one request.                  |
| <b>Request Guardrails</b>          | Limits request size and behavior.                               | Maximum 10 applications, 20 enterprise roles, 50 items, 1000-character comments. |

## Interview Question

**Q: What is the difference between Application-Level Settings and Global Settings in Saviynt?**

- **Application-Level Settings** apply to a **specific application** and define items such as its workflow and custom request form.
- **Global Settings** apply to **all access requests** across the system and control common behaviors, such as mandatory comments, account name entry, request limits, and start/end date options.

## Best Practices

- Use **OOTB connectors** whenever possible.
  - Always test provisioning in Development or UAT before Production.
  - Monitor provisioning queues regularly to identify bottlenecks.
  - Use dedicated **service accounts** with minimum required permissions.
  - Enable detailed logging for troubleshooting.
  - Configure retry policies for temporary API or network failures.
  - Validate successful provisioning through reconciliation jobs.
  - Periodically review failed tasks and resolve root causes.
-

# Common Provisioning Problems

| Issue                 | Possible Cause                       | Solution                          |
|-----------------------|--------------------------------------|-----------------------------------|
| Task stuck in Pending | Queue not processing                 | Check provisioning job            |
| Authentication failed | Invalid credentials                  | Verify service account            |
| Account not created   | Connector/API issue                  | Review connector logs             |
| Wrong role assigned   | Incorrect entitlement mapping        | Correct mapping configuration     |
| Duplicate account     | Correlation failure                  | Review identity correlation       |
| Manual task pending   | Administrator has not completed work | Follow up or configure escalation |

## Quick Revision

- **Provisioning** implements approved access changes.
- **Provisioning Tasks** contain the instructions required to perform those changes.
- Tasks support operations such as **Create, Modify, Enable, Disable, Delete, Assign Entitlement, Remove Entitlement, and Password Reset**.
- **Connected Applications** use connectors for automatic provisioning.
- **Disconnected Applications** rely on manual tasks or ServiceNow integration.
- The **Provisioning Queue** organizes task execution and improves scalability.
- **Validation** confirms successful execution before the task is completed.
- **Logs** and **Notifications** support monitoring, auditing, and troubleshooting.
- Always troubleshoot in the order: **Request** → **Task** → **Queue** → **Connector** → **Endpoint** → **Logs** → **Target Application**.



The content you shared is the **actual Saviynt University module**, which contains many implementation details that weren't covered in my previous summary. Below are **complete, office-evaluation-quality notes** covering **everything** from the slides in simple language with explanations, internal workflow, real examples, interview questions, and best practices.

---

## Course 6 – Access Request System (ARS)

### Module 6 – Intelligent Recommendations Using Dynamic Peer Groups

---

#### Introduction

Large organizations may have:

- 50,000+ employees
- Hundreds of applications
- Thousands of roles
- Thousands of entitlements

When a new employee joins, they usually don't know:

- Which application they need
- Which role is correct
- Which AD Group they should request
- Which SAP Role is required

This creates problems like:

- Wrong access requests
- Too many approvals
- Increased IT workload
- Security risks
- Compliance issues

To solve this problem, Saviynt introduced **Intelligent Recommendations** using **Dynamic Peer Groups**.

Instead of asking users to search through thousands of entitlements, Saviynt automatically recommends the most appropriate access based on similar employees.

---

# Evolution of Intelligence with Peer Groups

Saviynt's recommendation engine has evolved through three stages:

1. Manual Peer Groups
  2. Static Peer Groups
  3. Dynamic Peer Groups
- 

## 1. Manual Peer Group

### Definition

In Manual Peer Groups, administrators manually create groups and manually assign users to those groups.

Example:

Finance Team

↓

Administrator manually adds:

Rahul

Priya

John

Anita

Every time a new employee joins,  
the administrator must manually add them.

---

## Characteristics

- Completely manual
  - Time-consuming
  - Difficult to maintain
  - High administrative effort
  - Not suitable for large organizations
- 

## Example

Company has

5,000 employees.

Every month

200 employees join.

Administrator must manually assign each employee to the correct peer group.

Very inefficient.

---

## Advantages

- Full administrator control
  - Easy for very small organizations
- 

## Disadvantages

- Poor scalability
  - High maintenance
  - Human errors
  - Outdated quickly
-

## 2. Static Peer Group

### Definition

Static Peer Groups automatically include users based on predefined rules, but those rules remain fixed until an administrator changes them.

---

### Example

Rule:

Department = Investment Banking

Job Role = Financial Analyst

Manager = John Smith

Title = Second VP

Only users matching these conditions belong to the group.

---

### Characteristics

- Rule-based
  - Members fixed
  - Requires manual rule updates
  - Less flexible
- 

### Example

Finance Department renamed to:

Global Finance

Static rule still looks for:

Finance

Recommendations become incorrect until the rule is manually updated.

---

## Advantages

- Better than manual grouping
  - Easier maintenance
  - Consistent grouping
- 

## Disadvantages

- Cannot adapt automatically
  - Does not learn from user behavior
  - Can become outdated
- 

# 3. Dynamic Peer Group

## Definition

Dynamic Peer Groups are automatically created using multiple identity attributes and statistical models.

Membership changes automatically whenever user information changes.

---

## Example

Employee:

Department = Finance

Location = Hyderabad

Job Title = Financial Analyst

Manager = John

If tomorrow the employee moves to HR,

Saviynt automatically removes the employee from the Finance peer group and places them in the HR peer group.

No administrator action required.

---

## Characteristics

- AI driven
  - Automatic
  - Adaptive
  - Self-maintaining
  - Highly accurate
- 

## Why Dynamic Peer Groups are Better

Instead of using only one attribute,

Saviynt analyzes many attributes together.

Example:

Department

- 

Location

- 

Business Unit

- 

Manager

- 

Job Title

↓

Creates much more accurate recommendations.

---

## Comparison

| Manual             | Static               | Dynamic            |
|--------------------|----------------------|--------------------|
| Manual creation    | Rule-based           | AI-driven          |
| Manual maintenance | Manual rule updates  | Automatic updates  |
| High effort        | Medium effort        | Very low effort    |
| Poor scalability   | Good                 | Excellent          |
| Not intelligent    | Limited intelligence | Highly intelligent |

---

## Dynamic Peer Groups in Intelligent Recommendations

Dynamic Peer Groups automatically analyze user access across multiple predefined attributes.

They generate intelligent access recommendations based on similar users.

---

## Benefits

### 1. Automatically analyzes user access

Saviynt studies existing access across thousands of employees.

Instead of analyzing one user,

it analyzes the entire organization.

---

## **2. Reduces manual intervention**

Administrators no longer create access models manually.

The recommendation engine generates them automatically.

---

## **3. Reduces human errors**

Manual approvals often assign incorrect access.

AI recommendations reduce mistakes.

---

## **4. Auto-Correction**

The recommendation engine continuously learns.

If access patterns change,

recommendations automatically improve.

---

## **Example**

Last year

Finance users mostly required SAP.

This year

Finance users also require Oracle ERP.

The model automatically learns this new pattern.

---

## **5. Better Prediction Accuracy**

Recommendations become:



- More accurate
  - More reliable
  - More relevant
- 

## 6. Better Confidence

Every recommendation includes a confidence percentage.

Example

SAP Finance

98%

Oracle ERP

95%

AWS Admin

15%

Higher confidence indicates stronger recommendation.

---

## 7. Continuous Compliance

Because recommendations update automatically,

organizations remain compliant without continuously rebuilding peer groups.

---

## Most Critical Settings

This is one of the most important interview topics.

---

# 1. User Attributes for Recommendation Rules

Saviynt creates recommendations using user attributes.

Examples

- Department
  - Job Title
  - Business Unit
  - Cost Center
  - Manager
  - Employee Type
  - Location
  - Organization
- 

## Recommended Number

Use

5–10 attributes.

Maximum supported

40 attributes.

Too many attributes increase computation and reduce performance.

---

# 2. Confidence Threshold

## Definition

Confidence Threshold determines when Saviynt should display a recommendation.

---

## Recommended Value

30%–60%

---

### Example

Confidence

85%

↓

Show Recommendation

Confidence

15%

↓

Do Not Recommend

---

## 3. Minimum User Threshold

### Definition

Minimum number of similar users required before creating a recommendation.

---

## Recommended Value

5–15 users.

---

### Example

If only

2 users

have SAP Finance,

Saviynt may not recommend it.

If

20 users

have SAP Finance,

recommendation becomes reliable.

---

## Guidelines for Attribute Selection

---

### Select Key Attributes

Choose

7–8 meaningful attributes.

Examples

Department

Manager

Job Title

Location

Business Unit

Cost Center

Employee Type

Organization

---

## Avoid Overloading

Adding

30 or 40 attributes

does not improve accuracy.

Instead,

it increases computation time.

---

## Evaluate Effectiveness

Saviynt provides charts and reports showing:

- Which attributes improve recommendations
- Which attributes contribute very little

Administrators should remove unnecessary attributes.

---

## How to Enable Intelligent Recommendations

Navigate

Admin

↓

Global Configuration

↓

Intelligence

Enable

- Intelligent Recommendations
  - Trust Modelling
- 

## Why Enable Trust Modelling?

Trust Modelling improves recommendation quality by assigning confidence levels based on historical access patterns.

It helps approvers trust the recommendations.

---

## Recommendation Job

After enabling recommendations,

schedule the

Recommendations Job.

This automatically creates

Recommendation\_SystemDefault

Job Trigger.

---

## Recommendation\_SystemDefault Job

### Purpose

Generates:

- Dynamic Peer Groups
- Recommendation Rules
- Peer Analysis
- Access Models

Used by:

- Access Requests
  - Certifications
- 

## Important Points

This trigger name is reserved.

Do not rename it.

Do not schedule multiple conflicting recommendation jobs at the same time.

Otherwise,

recommendation generation may fail or produce inconsistent results.

---

## Creating Your Own Recommendation Job

Navigate

Admin

↓

Job Control Panel

↓

Add New Job

Choose

Job Type



Recommendations

Enable

Dynamic Recommendations

Choose

Analysis Type

Run Job

---

## Analysis Types

Analysis may include:

- Applications
- Roles
- Entitlements
- Accounts

depending on organizational requirements.

---

## Intelligent Recommendation Rulesets

After the job completes,

navigate

Recommendation Rulesets



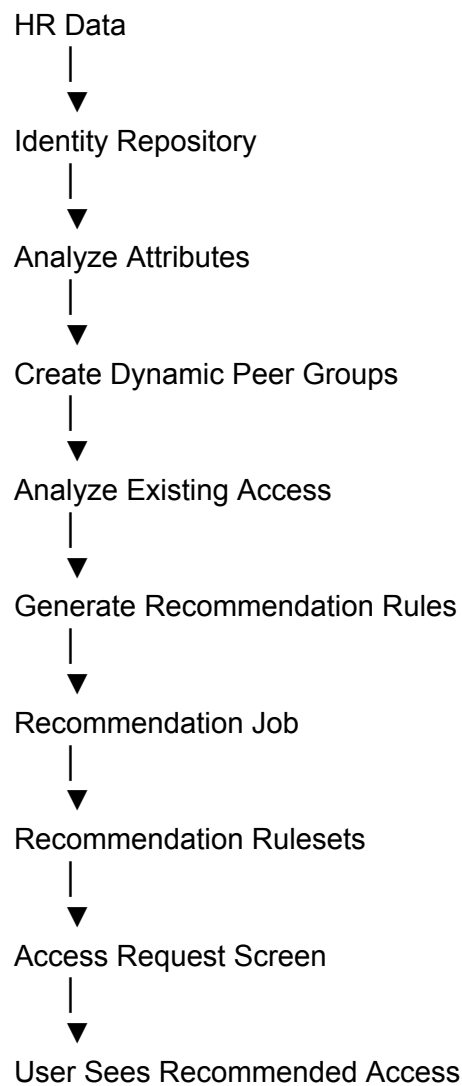
Intelligent Recommendations

Here administrators can view:



- Generated recommendation rules
  - Dynamic Peer Groups
  - Confidence Scores
  - Suggested access
- 

## Internal Workflow



## Real Project Example

ABC Bank

Employees

25,000

Applications

600

New employee

Rahul

Department

Finance

Location

Hyderabad

Job Title

Financial Analyst

Saviynt automatically finds

600 Finance Analysts

Most have

- SAP Finance
- Oracle ERP
- Finance SharePoint
- Microsoft Teams

These are automatically recommended.

Rahul simply selects them.

Approval process starts.

No searching required.

---

# Best Practices

- Use **Dynamic Peer Groups** instead of Manual or Static groups whenever possible.
  - Select **5–10 meaningful user attributes** for recommendation generation.
  - Avoid adding unnecessary attributes because they increase computation without improving accuracy.
  - Set the **Confidence Threshold** between **30% and 60%**.
  - Configure the **Minimum User Threshold** between **5 and 15** users.
  - Enable **Trust Modelling** to improve recommendation quality.
  - Schedule the **Recommendations Job** regularly so peer groups stay up to date.
  - Do **not** run multiple recommendation jobs simultaneously if they conflict.
  - Review generated recommendation rules periodically and remove outdated rules.
- 

## Interview Questions

### Q1. What are Intelligent Recommendations?

Intelligent Recommendations are AI-generated suggestions that recommend applications, roles, and entitlements based on access patterns of similar users.

---

### Q2. What is a Dynamic Peer Group?

A Dynamic Peer Group is an automatically maintained group of users with similar business attributes, used to generate accurate access recommendations.

---

### Q3. What is the difference between Manual, Static, and Dynamic Peer Groups?

- **Manual Peer Group:** Created and maintained manually.
  - **Static Peer Group:** Rule-based but requires manual updates when business rules change.
  - **Dynamic Peer Group:** Automatically updates using multiple attributes and statistical models.
-

#### **Q4. Why are Dynamic Peer Groups preferred?**

They automatically adapt to organizational changes, reduce manual work, improve recommendation accuracy, and support continuous compliance.

---

#### **Q5. What is the recommended number of attributes for recommendation generation?**

Saviynt recommends using **5–10 attributes**, with **7–8 meaningful attributes** often providing the best balance between accuracy and performance.

---

#### **Q6. What is the recommended Confidence Threshold?**

A **Confidence Threshold between 30% and 60%** is recommended to balance recommendation quality and coverage.

---

#### **Q7. What is the Minimum User Threshold?**

It is the minimum number of similar users required before Saviynt creates a recommendation rule. The recommended value is **5–15 users**.

---

#### **Q8. What is Trust Modelling?**

Trust Modelling evaluates historical access patterns and assigns confidence to recommendations, helping approvers make informed decisions.

---

#### **Q9. What is the purpose of the Recommendation\_SystemDefault job?**

It generates Dynamic Peer Groups, recommendation rules, and peer access insights used in **Access Requests** and **Access Certifications**.

---

#### **Q10. Where do you enable Intelligent Recommendations?**

Navigate to:

**Admin → Global Configuration → Intelligence**

Enable:

- Intelligent Recommendations
- Trust Modelling

Then schedule the **Recommendations Job**.

---

## Quick Revision

- **Manual Peer Groups** require manual creation and maintenance.
- **Static Peer Groups** use fixed rules but need manual updates.
- **Dynamic Peer Groups** automatically adapt using multiple user attributes and statistical models.
- Dynamic recommendations analyze **existing access patterns** to suggest the most appropriate access.
- Use **5–10 attributes** (ideally **7–8**) for optimal results.
- Recommended **Confidence Threshold: 30%–60%**.
- Recommended **Minimum User Threshold: 5–15** users.
- Enable **Intelligent Recommendations** and **Trust Modelling** under **Admin → Global Configuration → Intelligence**.
- Run the **Recommendations Job**, which creates the reserved **Recommendation\_SystemDefault** trigger.
- View generated recommendation rules under **Recommendation Rulesets → Intelligent Recommendations**.

Below are **complete office evaluation notes** for **Course 6 – Module 7: Recommendations in Access Request**. This is a continuation of **Dynamic Peer Groups**, but instead of explaining **how recommendations are generated**, this module explains **how they are actually shown to users and approvers during an Access Request**.

These notes cover **every topic from your slides**, along with implementation details, internal workflow, configuration, real project examples, best practices, troubleshooting, and interview questions.

---

# Module 7 – Recommendations in Access Request

---

## Introduction

Organizations often have:

- Thousands of users
- Hundreds of applications
- Thousands of roles
- Thousands of entitlements

When employees request access, they often don't know:

- Which application to request
- Which role is correct
- Which entitlement is required

Even managers sometimes don't know whether a request should be approved.

Saviynt solves this problem using **Recommendations in Access Requests**.

The recommendation engine analyzes users with similar job profiles and suggests the most appropriate applications, roles, and entitlements during the access request process.

---

## What are Recommendations in Access Requests?

### Definition

Recommendations in Access Requests are **AI-generated suggestions displayed while requesting access**.

These recommendations help users select the correct access and help approvers make informed approval decisions.

---

## Simple Definition

Saviynt recommends the best applications, roles, and permissions while the user is requesting access.

---

## Why are Recommendations Needed?

Without recommendations:

- Users guess which access to request.
- Wrong requests increase.
- Managers reject more requests.
- IT spends more time correcting requests.
- Security risks increase.

With recommendations:

- Users request the correct access.
  - Approval becomes easier.
  - Security improves.
  - Compliance improves.
- 

## Recommendation Engine

Saviynt uses:

- Machine Learning
- Dynamic Peer Groups
- Historical Access Data
- Statistical Models

to recommend:

- Applications
- Roles
- Entitlements



---

# Proprietary Machine Learning Engine

Saviynt uses its own proprietary Machine Learning algorithms.

The engine analyzes:

- User identity
- Department
- Job Title
- Manager
- Existing access
- Peer access
- Historical approvals

Unlike rule-based systems,

Machine Learning continuously improves recommendations.

---

## What Does the Engine Recommend?

The engine recommends:

- Applications
  - Business Roles
  - Technical Roles
  - Entitlements
  - Groups
  - Permissions
- 

## Inlier Detection

This is an important interview topic.

### Definition

**Inlier Detection** identifies access that is common among users in the same peer group.

Instead of looking for unusual access,

it looks for access that most similar employees already have.

---

## Example

Finance Analysts:

- SAP Finance
- Oracle ERP
- Teams
- Finance SharePoint

95% of Finance Analysts have these applications.

Saviynt recommends the same access for new Finance Analysts.

---

## Why Inlier Detection?

Because common access is usually the correct access.

It reduces:

- Wrong requests
  - Security risks
  - Manual approvals
- 

## Transparent Recommendations

Saviynt doesn't simply recommend access.

It also explains **why**.

Example

Recommended:

SAP Finance

Reason:

92% of Finance Analysts in Hyderabad already have this application.

This improves approver confidence.

---

## One-Click Recommendation

Users don't need to search manually.

They simply click the recommended application.

This improves:

- Speed
  - User experience
  - Accuracy
- 

## Recommendations Available In

Recommendations are displayed during:

### Request Access for Self

User requests access for themselves.

Example

Rahul requests SAP.

Recommendations appear automatically.

---

## Request Access for Others

Managers request access for employees.

Recommendations are generated based on the employee's identity—not the manager's.

---

## Recommendation Workflow

Employee Opens Access Request



Identity Loaded



Dynamic Peer Group Identified



Machine Learning Analysis



Recommendation Engine



Recommended Applications  
Recommended Roles  
Recommended Entitlements



User Selects Access



Approval Workflow

---

## How Recommendations Work

The process follows six major steps.

---

## Step 1

Enable Intelligent Recommendations.

---

## Step 2

Run Recommendations Job.

---

## Step 3

Dynamic Peer Groups are generated.

---

## Step 4

Recommendation Model is built.

---

## Step 5

Model is measured and configured.

---

## Step 6

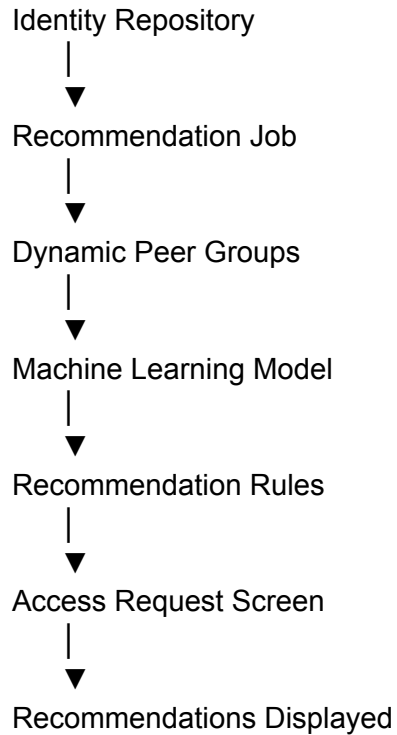
Recommendations appear during Access Requests.

---

## Internal Workflow

HR System





## Measurable and Configurable Model

Saviynt's recommendation engine is configurable.

Administrators can adjust:

- Attributes
- Confidence Threshold
- Minimum User Threshold
- Recommendation Rules

This helps improve recommendation quality.

---

## Enable Intelligent Recommendations

Navigate to

Admin



Global Configuration



Intelligence

Enable

- Intelligent Recommendations
  - Include Roles in Intelligent Recommendations
  - Trust Modelling
- 

## Why Include Roles?

Normally,

Saviynt recommends only applications.

If

Include Roles

is enabled,

Saviynt also recommends:

- Business Roles
- Technical Roles

instead of only applications.

---

## Trust Modelling

### Definition

Trust Modelling calculates confidence for recommendations.

Example

Recommendation

SAP Finance

Confidence

97%

This tells approvers how reliable the recommendation is.

---

## Schedule Recommendation Job

After enabling recommendations,

schedule

Recommendations Job.

This creates

Recommendation\_SystemDefault

Job Trigger.

---

## Recommendation\_SystemDefault

Purpose

Generates:

- Dynamic Peer Groups
- Recommendation Rules
- Confidence Scores

Used in:



- Access Requests
  - Certifications
- 

# Configuration for Access Requests

Navigate

Admin



Global Configuration



Intelligence

Enable

## Provide Recommendations for Access Requests

Displays application recommendations.

---

## Provide Role Recommendations for Access Requests

Displays Business Role recommendations.

---

## Provide Approval Confidence for Access Requests

Displays confidence percentages to approvers.

Example

SAP Finance

Confidence

96%

---

## Recommendations During Access Review

Recommendations are also available during:

Access Certification

Access Review

Before recommendations appear,

run

Recommendations Job

↓

Trust Modelling Job

---

## Why Trust Modelling Job?

Trust Modelling calculates:

- Confidence
- Trust Score
- Recommendation Reliability

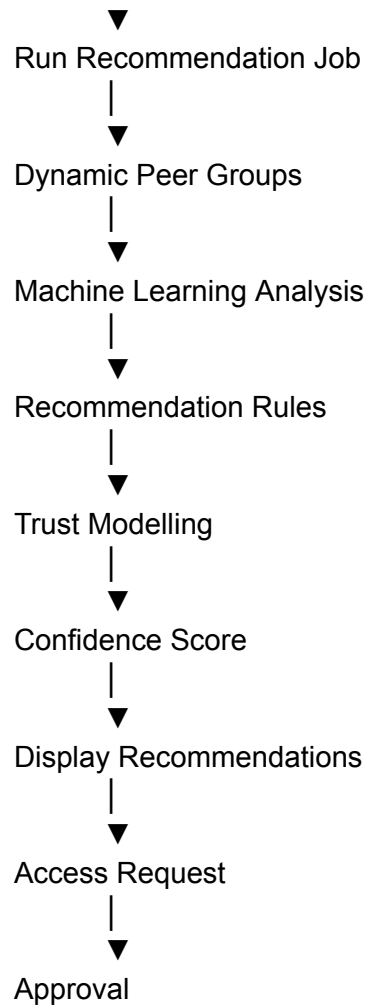
Approvers can decide more confidently.

---

## Complete Recommendation Workflow

Enable Recommendations

|



## Real Project Example

ABC Bank

Employees

30,000

Applications

750

Rahul joins Finance.

Identity

Department

Finance

Location

Hyderabad

Job

Financial Analyst

Recommendation Engine finds

950 Finance Analysts.

Most have

- SAP Finance
- Oracle ERP
- Finance SharePoint
- Teams

Recommendations shown

Confidence

SAP Finance

98%

Oracle ERP

95%

Finance SharePoint

94%

Rahul selects all.

Manager immediately understands the request is normal because confidence is high.

Approval completed.

---

# Benefits

- Better user experience
  - Faster requests
  - Faster approvals
  - Better compliance
  - Standardized access
  - Lower IT workload
  - Higher prediction accuracy
  - Reduced human error
  - Better security
- 

# Best Practices

- Enable **Intelligent Recommendations**, **Role Recommendations**, and **Trust Modelling** together.
  - Run the **Recommendations Job** regularly to keep peer groups current.
  - Run the **Trust Modelling Job** after recommendation generation to update confidence scores.
  - Use **Dynamic Peer Groups** instead of manually maintained groups.
  - Monitor confidence scores and recommendation accuracy.
  - Review recommendation rules periodically and remove obsolete rules.
  - Ensure HR identity attributes (department, title, manager, location) are accurate because recommendation quality depends on them.
- 

# Common Issues

## Recommendations Not Displayed

### Possible Causes

- Intelligent Recommendations disabled.
- Recommendation Job not run.
- Dynamic Peer Groups not generated.

### Resolution

Enable recommendations and rerun the Recommendation Job.

---

## **Role Recommendations Missing**

### **Cause**

"Include Roles in Intelligent Recommendations" is disabled.

### **Resolution**

Enable the setting and rerun the Recommendation Job.

---

## **Confidence Score Not Displayed**

### **Cause**

Trust Modelling is not enabled or the Trust Modelling Job has not run.

### **Resolution**

Enable Trust Modelling and execute the Trust Modelling Job.

---

## **Incorrect Recommendations**

### **Cause**

- Incorrect HR attributes.
- Outdated peer groups.
- Insufficient historical access data.

### **Resolution**

Review identity data, regenerate Dynamic Peer Groups, and rerun recommendation jobs.

---

# **Interview Questions**

## **Q1. What are Recommendations in Access Requests?**

Recommendations are AI-generated suggestions displayed during the access request process to help users request the correct applications, roles, and entitlements.

---

## **Q2. Which technologies are used to generate recommendations?**

Saviynt uses:

- Machine Learning
  - Dynamic Peer Groups
  - Historical Access Data
  - Statistical Models
- 

## **Q3. What is Inlier Detection?**

Inlier Detection identifies access commonly assigned to users within the same peer group and recommends those permissions.

---

## **Q4. Where are recommendations displayed?**

Recommendations are available in:

- **Request Access for Self**
  - **Request Access for Others**
  - **Access Reviews/Certifications** (after appropriate jobs are run)
- 

## **Q5. What is the purpose of the Recommendations Job?**

The Recommendations Job generates Dynamic Peer Groups, recommendation rules, and the recommendation model used during access requests.

---

## **Q6. Why is Trust Modelling required?**

Trust Modelling calculates confidence scores that help approvers understand how reliable each recommendation is.

---

### Q7. What is the purpose of "Provide Approval Confidence for Access Requests"?

It displays confidence percentages for recommendations, allowing approvers to make better-informed decisions.

---

### Q8. What does "Include Roles in Intelligent Recommendations" do?

It enables Saviynt to recommend Business Roles and Technical Roles in addition to applications and entitlements.

---

### Q9. Can recommendations automatically grant access?

No. Recommendations only suggest appropriate access. The normal approval workflow and provisioning process must still be completed.

---

### Q10. What are the benefits of recommendations in access requests?

- Faster requests
  - Better approval decisions
  - Improved compliance
  - Reduced incorrect requests
  - Lower IT workload
  - Better user experience
  - Standardized access across similar users
- 

## Quick Revision

- **Recommendations in Access Requests** provide AI-based suggestions during access requests.
- The recommendation engine uses **Machine Learning**, **Dynamic Peer Groups**, and **Historical Access Patterns**.
- **Inlier Detection** recommends access commonly used by similar employees.
- Recommendations are available in **Request for Self** and **Request for Others** workflows.



- Enable:
  - **Intelligent Recommendations**
  - **Include Roles in Intelligent Recommendations**
  - **Trust Modelling**
- Run the **Recommendations Job** to generate Dynamic Peer Groups and recommendation rules.
- Run the **Trust Modelling Job** to generate confidence scores.
- Enable:
  - **Provide Recommendations for Access Requests**
  - **Provide Role Recommendations for Access Requests**
  - **Provide Approval Confidence for Access Requests**
- Recommendations improve accuracy but **do not replace approvals**; every request still follows the configured approval workflow before provisioning.

# Module 8 – Email Notifications and Templates

---

## Introduction

In Saviynt, every important activity generates notifications.

Examples:

- User requests access
- Manager approval required
- Request approved
- Request rejected
- Provisioning completed
- Password reset
- Certification launched

Instead of manually informing users,

Saviynt automatically sends emails using **Email Templates**.

---

## What are Email Notifications?

### Definition

**Email Notifications** are automatically generated messages sent by Saviynt to users, managers, application owners, administrators, or other stakeholders when specific events occur.

---

### Simple Definition

**Email Notifications keep users informed about every important action happening in Saviynt.**

---

# Why are Email Notifications Required?

Without notifications:

- Users don't know request status.
- Managers miss approval requests.
- IT receives many support calls.
- Users don't know when access is granted.
- Compliance tracking becomes difficult.

Notifications solve these problems.

---

## Benefits of Email Notifications

- Automatic communication
  - Faster approvals
  - Better user experience
  - Reduced Help Desk calls
  - Audit support
  - Improved compliance
  - Timely updates
- 

## What is an Email Template?

### Definition

An **Email Template** is a predefined email format used by Saviynt to send consistent notifications.

Instead of creating emails every time,

Saviynt fills in dynamic information and sends the message automatically.

---

## Simple Definition

An Email Template is a reusable email format with placeholders for user-specific information.

---

## Components of an Email Template

An email template generally contains:

- Subject
  - Greeting
  - Message Body
  - Dynamic Variables
  - Footer
- 

## Example Email Template

Subject:

Access Request Approved

Hello \${username},

Your request for Salesforce has been approved.

Application : Salesforce

Requested By : Rahul

Status : Approved

Thank You,  
Saviynt Team

`${username}` is a **dynamic variable**. Saviynt replaces it with the actual user's name before sending the email.

---

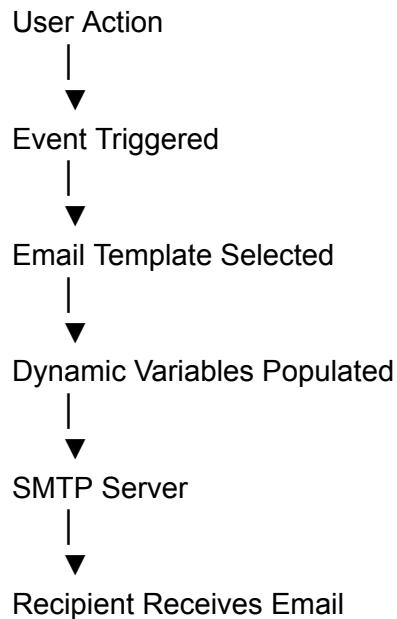
# Dynamic Variables

Dynamic Variables are placeholders replaced with actual values at runtime.

Examples:

- \${username}
  - \${firstname}
  - \${lastname}
  - \${application}
  - \${requestid}
  - \${manager}
  - \${status}
  - \${approvaldate}
- 

# Email Notification Workflow



# Common Notification Events

Saviynt sends emails for many events.

---

## 1. Access Request Submitted

Sent to:

- Requester

Purpose:

Confirms request submission.

---

## 2. Approval Required

Sent to:

- Manager
- Application Owner
- Resource Owner

Purpose:

Requests approval.

---

## 3. Request Approved

Sent to:

Requester

Purpose:

Informs that approval is complete.

---

## 4. Request Rejected

Sent to:

Requester

Purpose:

Explains that access was rejected.

---

## 5. Provisioning Started

Sent to:

Administrator (optional)

Purpose:

Provisioning has begun.

---

## 6. Provisioning Completed

Sent to:

Requester

Purpose:

Access is now available.

---

## 7. Provisioning Failed

Sent to:

Administrator

Purpose:

Immediate troubleshooting.

---

## 8. Password Reset

Sent to:

User

Purpose:

Password reset confirmation.

---

## 9. Account Locked

Sent to:

User

Purpose:

Account security notification.

---

## 10. Certification Notifications

Sent to:

- Reviewer
- Manager
- Campaign Owner

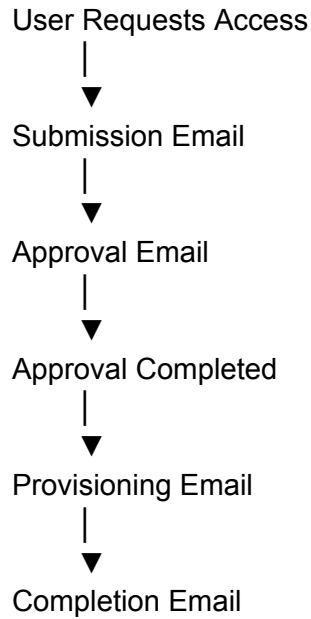
Purpose:

Inform users about certification campaigns.

---

# Email Notification Flow During Access Request





# SMTP Server

## What is SMTP?

**SMTP (Simple Mail Transfer Protocol)** is the protocol used to send emails.

Saviynt uses an SMTP server to deliver notification emails.

---

## SMTP Configuration

Administrators configure:

- SMTP Host
  - Port
  - Username
  - Password
  - TLS/SSL
  - Sender Email Address
-

# HTML Email Templates

Saviynt supports HTML templates.

Advantages:

- Better formatting
  - Company logo
  - Colors
  - Tables
  - Hyperlinks
- 

## Plain Text Templates

Simple text emails.

Advantages:

- Faster
  - Lightweight
  - Compatible with all email clients
- 

## Email Template Best Practices

A good template should:

- Be short.
  - Clearly explain the action.
  - Include request details.
  - Use meaningful subject lines.
  - Avoid unnecessary information.
  - Include support contact details if required.
- 

## Real Project Example

ABC Bank

Employee:

Rahul

Requests Salesforce access.

### **Step 1**

Request submitted.

↓

Saviynt sends confirmation email.

---

### **Step 2**

Manager receives approval email.

---

### **Step 3**

Manager approves.

↓

Rahul receives approval email.

---

### **Step 4**

Provisioning completed.

↓

Rahul receives:

"Salesforce account successfully created."

---

### **Step 5**

Audit logs updated.

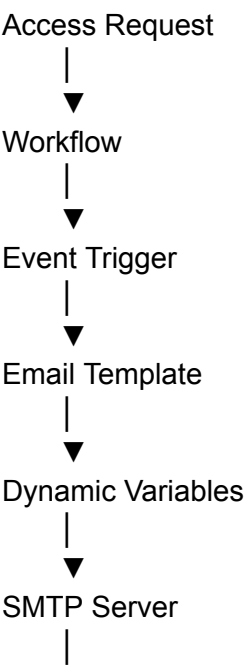
---

# Notification Types

| Event                  | Recipient     |
|------------------------|---------------|
| Request Submitted      | Requester     |
| Approval Required      | Approver      |
| Request Approved       | Requester     |
| Request Rejected       | Requester     |
| Provisioning Completed | Requester     |
| Provisioning Failed    | Administrator |
| Password Reset         | User          |
| Certification          | Reviewer      |

---

# Internal Architecture



▼  
Recipient

---

## Common Problems

### Emails Not Sent

Possible Causes:

- SMTP not configured
  - SMTP server unavailable
  - Firewall issue
  - Wrong credentials
- 

### Wrong Recipient

Cause:

Incorrect email mapping.

---

### Empty Dynamic Variables

Cause:

Incorrect template variables.

---

### Duplicate Emails

Cause:

Multiple notification triggers.

---

# Delayed Emails

Cause:

SMTP queue delay.

---

## Troubleshooting

### Step 1

Check SMTP configuration.

---

### Step 2

Verify SMTP credentials.

---

### Step 3

Check firewall/network connectivity.

---

### Step 4

Review email template configuration.

---

### Step 5

Verify dynamic variables.

---

### Step 6

Check notification logs.

---

## Step 7

Send test email.

---

## Best Practices

- Use descriptive email subjects.
  - Keep templates simple and professional.
  - Use dynamic variables instead of hard-coded values.
  - Test templates before production.
  - Configure SMTP with TLS/SSL.
  - Avoid sending unnecessary notifications.
  - Review templates regularly.
  - Use HTML only when required.
  - Maintain separate templates for different events.
- 

## Interview Questions

### Q1. What are Email Notifications in Saviynt?

Email Notifications are automatically generated emails sent when predefined events such as access requests, approvals, provisioning, or certifications occur.

---

### Q2. What is an Email Template?

An Email Template is a reusable email format that contains fixed text and dynamic variables used to send consistent notifications.

---

### Q3. What are Dynamic Variables?

Dynamic Variables are placeholders like `${username}` or `${application}` that Saviynt replaces with actual values before sending an email.

---

#### **Q4. What protocol does Saviynt use to send emails?**

Saviynt uses **SMTP (Simple Mail Transfer Protocol)** to send notification emails.

---

#### **Q5. When are Email Notifications sent?**

Common events include:

- Request Submitted
  - Approval Required
  - Request Approved
  - Request Rejected
  - Provisioning Completed
  - Provisioning Failed
  - Password Reset
  - Certification Campaigns
- 

#### **Q6. What information is typically included in an approval email?**

It usually contains the requester, application name, requested access, request ID, and an approval link or instructions.

---

#### **Q7. What are common reasons emails are not sent?**

- SMTP configuration issues
  - Invalid credentials
  - Network/firewall problems
  - Incorrect recipient mapping
  - Email template configuration errors
- 

#### **Q8. Why should dynamic variables be used instead of hard-coded values?**

Dynamic variables automatically personalize emails and make templates reusable for different users and requests.



---

### Q9. What are the advantages of HTML email templates?

They support branding, formatting, tables, hyperlinks, and logos, providing a more professional appearance.

---

### Q10. How do you troubleshoot email notification issues?

Check SMTP settings, verify credentials, test network connectivity, review email templates, validate dynamic variables, inspect notification logs, and send a test email.

---

## Quick Revision

- **Email Notifications** automatically inform users about important Saviynt events.
- **Email Templates** provide reusable email formats with dynamic placeholders.
- **Dynamic Variables** (e.g., `${username}`, `${application}`) are replaced with real values at runtime.
- Notifications are commonly sent for **Access Requests**, **Approvals**, **Provisioning**, **Password Resets**, and **Certifications**.
- Saviynt sends emails using an **SMTP server**.
- Templates can be **Plain Text** or **HTML**.
- Always test templates before production and keep them concise.
- Troubleshoot by checking **SMTP configuration**, **credentials**, **network connectivity**, **template variables**, and **notification logs**.

# Module 9 – Introduction to JITAM (Just-In-Time Access Management)

---

## Introduction

In traditional Identity and Access Management (IAM), users often receive **permanent privileged access** such as:

- Database Administrator
- Windows Administrator
- AWS Administrator
- SAP Basis Admin

- Domain Administrator

Keeping privileged access permanently creates security risks.

To reduce these risks, Saviynt provides **Just-In-Time Access Management (JITAM)**.

Instead of granting privileged access permanently, JITAM provides **temporary access only when it is needed**.

---

# What is JITAM?

## Definition

**Just-In-Time Access Management (JITAM)** is a security feature that grants users temporary privileged access for a specific task or duration. Once the approved time expires, the access is automatically removed.

---

## Simple Definition

**JITAM gives users privileged access only for a short time and automatically removes it afterward.**

---

# Why is JITAM Required?

Many users only need privileged access occasionally.

For example:

- Server maintenance
- Production deployment
- Database patching
- Emergency troubleshooting

Granting permanent administrator access for these occasional tasks increases the attack surface and violates the **Principle of Least Privilege**.

JITAM solves this problem by providing access only when required.

---

## Problems with Permanent Access

Permanent administrator access can lead to:

- Insider threats
- Accidental changes
- Data leakage
- Unauthorized activity
- Privilege misuse
- Compliance violations

Example:

A Database Administrator has permanent production access but only uses it once a month.

For the remaining days, the privileged account is unnecessary and increases security risk.

---

## How JITAM Solves the Problem

Instead of assigning permanent access:

- User requests privileged access.
  - Manager or Application Owner approves.
  - Access is granted temporarily.
  - Access automatically expires after the approved duration.
- 

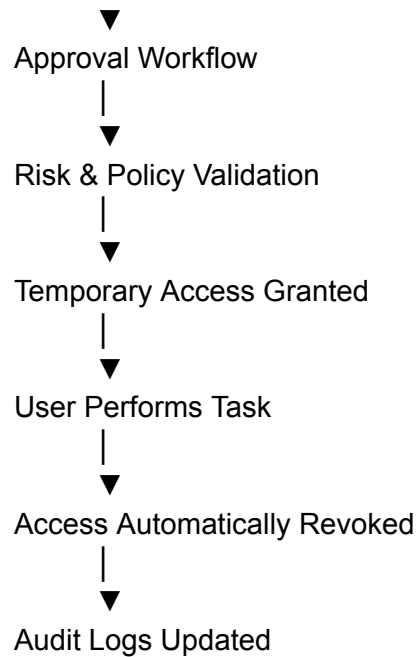
## JITAM Workflow

Need Privileged Access



Submit JIT Request





## Components of JITAM

A typical JITAM process includes:

- User
- Access Request
- Approval Workflow
- Policy Validation
- Provisioning
- Time-Based Expiry
- Automatic Deprovisioning
- Audit Logs

---

## JIT Access Lifecycle

Request



Approval

↓

Provisioning

↓

Temporary Usage

↓

Automatic Expiration

↓

Deprovisioning

↓

Audit Complete

---

## Example Scenario

Rahul is a Database Engineer.

Normally he does **not** have Database Administrator privileges.

One weekend a production database patch must be installed.

Rahul:

- Requests Database Administrator access.
- Manager approves.
- DBA Team approves.
- Saviynt grants access for **4 hours**.
- Rahul completes the patch.
- After 4 hours Saviynt automatically removes the Database Administrator role.

No manual cleanup is required.

---

# Time-Based Access

JITAM always uses **time-limited access**.

Example:

| Access            | Duration   |
|-------------------|------------|
| AWS Admin         | 2 Hours    |
| Database Admin    | 4 Hours    |
| VPN Access        | 1 Day      |
| Production Server | 30 Minutes |

Once the time expires, Saviynt automatically removes the access.

---

# Approval Workflow

JIT requests usually require stricter approvals than normal access requests.

Example:

User

↓

Manager

↓

Application Owner

↓

Information Security

↓

Temporary Access Granted

High-risk requests may require additional approvals.

---

## Automatic Deprovisioning

One of the biggest advantages of JITAM is **automatic deprovisioning**.

When the approved duration ends:

- Roles are removed.
- Groups are removed.
- Accounts can be disabled if required.
- Audit logs are updated.

This prevents users from retaining unnecessary privileged access.

---

## Zero Standing Privileges (ZSP)

### Definition

Zero Standing Privileges means users **do not permanently possess privileged access**.

They receive administrator privileges only when needed.

JITAM is one of the primary methods used to implement Zero Standing Privileges.

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## Principle of Least Privilege (PoLP)

JITAM supports the **Principle of Least Privilege**.

Users receive:

- Minimum permissions



- Only for the required task
- Only for the approved duration

This significantly reduces security risks.

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## Benefits of JITAM

- Improves security
  - Reduces insider threats
  - Reduces privileged account misuse
  - Supports compliance
  - Automatic access removal
  - Better auditability
  - Lower attack surface
  - Supports Zero Standing Privileges
  - Reduces manual administration
- 

## Difference Between Permanent Access and JIT Access

| Permanent Access        | JIT Access                   |
|-------------------------|------------------------------|
| Always available        | Available only when approved |
| Higher security risk    | Lower security risk          |
| Manual removal          | Automatic removal            |
| Long-term privileges    | Temporary privileges         |
| Higher chance of misuse | Reduced chance of misuse     |

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## Real Project Example

**Company:** ABC Bank

Privileged Role:

AWS Administrator

Developers normally do not have administrator privileges.

A production deployment is scheduled for Saturday.

Workflow:

1. Developer submits a JIT request.
  2. Manager approves.
  3. Cloud Team approves.
  4. Saviynt provisions AWS Administrator role.
  5. Access remains active for **3 hours**.
  6. Deployment completes.
  7. Saviynt automatically removes the administrator role.
  8. Audit logs record every action.
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## Complete Internal Workflow

User Needs Admin Access



Submit JIT Request



Workflow Starts



Manager Approval



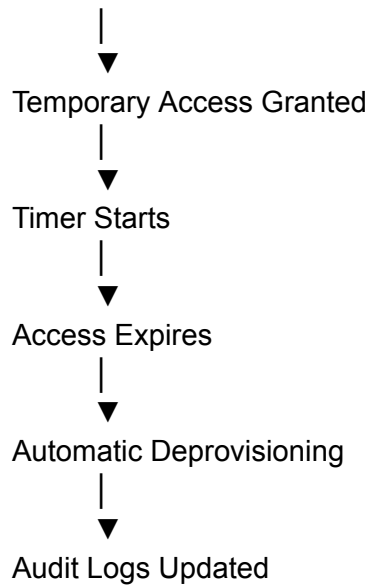
Application Owner Approval



Security Validation



Provisioning Task Created



## Best Practices

- Use JITAM for all privileged accounts whenever possible.
  - Require business justification for every JIT request.
  - Configure start and end times carefully.
  - Use multi-level approvals for high-risk access.
  - Enable automatic deprovisioning.
  - Monitor privileged sessions and audit logs.
  - Review JIT usage regularly.
  - Avoid granting permanent administrator roles unless absolutely necessary.
  - Apply the Principle of Least Privilege.
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## Common Issues

### Access Not Removed

#### Cause:

- Deprovisioning job failed.
- End time not configured.
- Workflow issue.

**Resolution:**

Verify deprovisioning jobs and expiry settings.

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**JIT Request Stuck****Cause:**

- Approval pending.
- Workflow misconfiguration.

**Resolution:**

Review workflow and approver assignments.

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**User Cannot Request JIT Access****Cause:**

- Role not marked as requestable.
- Incorrect request configuration.

**Resolution:**

Verify Request Option and Access Catalog configuration.

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**Incorrect Access Duration****Cause:**

- Wrong start or end time.
- Policy misconfiguration.

**Resolution:**

Review JIT policy and request details.

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# Troubleshooting Steps

1. Verify the JIT request was submitted successfully.
  2. Check approval workflow status.
  3. Confirm provisioning task creation.
  4. Validate connector or manual provisioning.
  5. Verify start and end times.
  6. Check deprovisioning jobs.
  7. Review audit logs and notifications.
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## Interview Questions

### Q1. What is JITAM?

JITAM (Just-In-Time Access Management) grants users temporary privileged access for a limited duration and automatically removes it when the approved time expires.

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### Q2. Why is JITAM important?

It reduces permanent privileged access, improves security, supports compliance, and minimizes insider threats.

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### Q3. What is the difference between permanent access and JIT access?

Permanent access is always available until manually removed, whereas JIT access is granted temporarily after approval and automatically revoked after the approved duration.

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### Q4. What is Zero Standing Privileges (ZSP)?

Zero Standing Privileges means users do not permanently hold privileged access. They receive elevated permissions only when required through mechanisms like JITAM.

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## **Q5. How does JITAM support the Principle of Least Privilege?**

It grants only the minimum required privileged access for a limited time, reducing unnecessary permissions.

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## **Q6. What happens when the JIT access duration expires?**

Saviynt automatically deprovisions the temporary access, updates audit logs, and restores the user's normal permission level.

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## **Q7. What types of access are commonly managed using JITAM?**

Examples include:

- Database Administrator
  - AWS Administrator
  - Windows Administrator
  - Domain Administrator
  - Production Server Access
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## **Q8. Why is automatic deprovisioning important?**

It ensures temporary privileged access is removed on time, reducing security risks and eliminating the need for manual cleanup.

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## **Q9. What approvals are typically required for JIT access?**

Manager approval, Application Owner approval, and sometimes Security or Compliance approval, depending on the organization's policies.

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## **Q10. What are the benefits of JITAM?**

- Improved security
- Reduced attack surface
- Better compliance

- Automatic access removal
  - Stronger audit trails
  - Support for Zero Standing Privileges
- 

## Quick Revision

- **JITAM (Just-In-Time Access Management)** provides **temporary privileged access**.
- It is commonly used for **administrator-level access** such as AWS Admin, Database Admin, or Production Server access.
- JIT requests follow an **approval workflow** before provisioning.
- Access is granted only for a **defined time period**.
- After the timer expires, **automatic deprovisioning** removes the access.
- JITAM supports **Zero Standing Privileges (ZSP)** and the **Principle of Least Privilege (PoLP)**.
- It improves security, reduces insider threats, and strengthens compliance by eliminating unnecessary permanent privileged access.

# Module 10 – Best Practices & Troubleshooting

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## Introduction

Even if Access Requests are configured correctly, issues can still occur because of:

- Incorrect workflows
- Missing approvals
- Connector failures
- Provisioning errors
- Notification failures
- Wrong mappings
- Recommendation issues

To reduce these issues, organizations follow **Best Practices** and use a structured **Troubleshooting approach**.

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## Best Practices

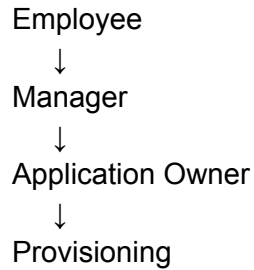
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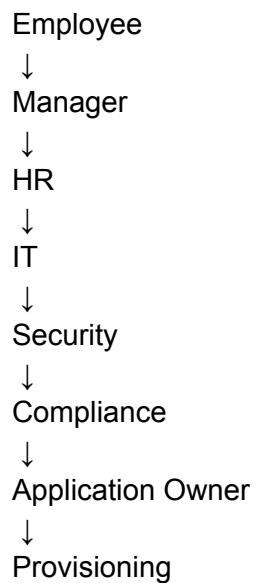
# 1. Design Simple Approval Workflows

Avoid unnecessary approval levels.

## Good Workflow



## Bad Workflow



Too many approval levels increase delays.

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# 2. Use Standard Workflows

Create reusable workflows instead of creating a new workflow for every application.

## Example

One Manager Approval workflow can be reused for:

- Salesforce
- ServiceNow
- Jira
- Concur

Benefits:

- Easy maintenance
  - Consistent approvals
  - Less configuration effort
- 

### 3. Enable Business Justification

Always require users to explain why they need access.

Example:

"I need SAP Finance access to work on the quarterly financial closing process."

Benefits:

- Better approval decisions
  - Audit evidence
  - Compliance support
- 

### 4. Use Start Date and End Date

For temporary access:

- Start Date controls when access begins.
- End Date controls when access is removed.

Example:

VPN Access

Start:

1 July

End:

10 July

Saviynt automatically provisions and later deprovisions the access.

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## 5. Use Temporary Access for Privileged Roles

Instead of permanently assigning:

- AWS Administrator
- Database Administrator
- Domain Administrator

Grant temporary access using **JITAM**.

Benefits:

- Reduced security risk
  - Supports Least Privilege
  - Automatic access removal
- 

## 6. Enable SoD (Segregation of Duties)

Prevent users from receiving conflicting access.

Example:

Do **not** allow the same user to have:

- Create Vendor
- Approve Payment

Benefits:

- Prevents fraud
  - Improves compliance
- 

## 7. Keep Access Catalog Clean

Remove:

- Obsolete applications
- Unused roles
- Duplicate entitlements

Benefits:

- Easier user experience
  - Better recommendations
  - Faster search
- 

## 8. Configure Notifications

Send emails for:

- Request Submitted
- Approval Required
- Request Approved
- Request Rejected
- Provisioning Completed
- Provisioning Failed

Benefits:

- Better communication
  - Fewer Help Desk tickets
- 

## 9. Use Dynamic Peer Groups

Instead of manually assigning peer groups,

use Dynamic Peer Groups for Intelligent Recommendations.

Benefits:

- Better recommendations
  - Less maintenance
  - Automatic updates
- 

## 10. Monitor Jobs Regularly

Monitor important scheduled jobs:

- Recommendations Job
- Trust Modelling Job
- Provisioning Jobs
- Import Jobs
- Reconciliation Jobs

Benefits:

- Prevents delays
  - Detects failures early
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## Common Access Request Issues

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### 1. Application Not Visible in Access Catalog

**Possible Causes**

- Request Option disabled
- Application not requestable

- Security System disabled
- Endpoint inactive

## **Resolution**

Verify:

- Request Option
  - Endpoint status
  - Security System configuration
- 

## **2. Workflow Not Triggered**

### **Possible Causes**

- Workflow not assigned
- Invalid workflow
- Incorrect approval configuration

### **Resolution**

Check:

- Security System
  - Workflow assignment
  - Approval rules
- 

## **3. Wrong Approver**

### **Possible Causes**

- Manager mapping incorrect
- Workflow configuration incorrect
- Resource owner missing

### **Resolution**

Verify:

- Manager relationship
  - Resource owner
  - Workflow configuration
- 

## 4. Approval Pending Forever

### Possible Causes

- Approver unavailable
- Escalation not configured
- Delegation not configured

### Resolution

Enable:

- Escalation
  - Delegation
  - Approval reminders
- 

## Provisioning Issues

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## Task Stuck in Pending

### Possible Causes

- Provisioning job not running
- Queue backlog
- Connector unavailable

### Resolution

Check:

- Provisioning Queue
  - Scheduled jobs
  - Connector status
- 

# Provisioning Failed

## Possible Causes

- Authentication failure
- Invalid credentials
- API failure
- Endpoint unavailable

## Resolution

Review:

- Connector logs
  - Endpoint
  - Authentication
  - API response
- 

# Wrong Role Assigned

## Possible Causes

- Incorrect entitlement mapping
- Wrong business role mapping

## Resolution

Review:

- Role mappings
  - Entitlement mappings
  - Provisioning rules
-



# Recommendations Issues

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## Recommendations Not Displayed

### Possible Causes

- Recommendation Job not executed
- Intelligent Recommendations disabled
- Dynamic Peer Groups missing

### Resolution

Run:

- Recommendations Job

Enable:

- Intelligent Recommendations
- 

## Confidence Score Missing

### Cause

Trust Modelling disabled.

### Resolution

Enable Trust Modelling.

Run Trust Modelling Job.

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## Wrong Recommendations

## Cause

Incorrect HR attributes.

## Resolution

Verify:

- Department
- Job Title
- Location
- Business Unit

Regenerate Dynamic Peer Groups.

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# Notification Issues

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## Emails Not Sent

### Possible Causes

- SMTP configuration incorrect
- Mail server unavailable
- Firewall issue

### Resolution

Check:

- SMTP Host
  - Port
  - Username
  - Password
  - Network connectivity
-

# Wrong Recipient

## Cause

Incorrect email mapping.

## Resolution

Verify:

- User email
  - Manager email
  - Template configuration
- 

# Duplicate Emails

## Cause

Multiple notification triggers.

## Resolution

Review notification configuration and remove duplicate triggers.

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# JITAM Issues

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# Access Not Removed

## Cause

- End date not configured
- Deprovisioning job failed

## Resolution

Verify:

- End Date
  - Deprovisioning Job
  - Provisioning Task status
- 

# JIT Request Not Working

## Cause

- Workflow missing
- Role not requestable

## Resolution

Check:

- Request Option
  - Workflow assignment
  - JIT configuration
- 

# Intelligent Recommendation Issues

| Problem                      | Cause                      | Solution                                  |
|------------------------------|----------------------------|-------------------------------------------|
| No recommendations           | Recommendation Job not run | Run Recommendation Job                    |
| Wrong recommendations        | Incorrect HR data          | Update identity attributes                |
| Low confidence               | Too little historical data | Allow model to learn or adjust thresholds |
| Role recommendations missing | Include Roles disabled     | Enable role recommendations               |

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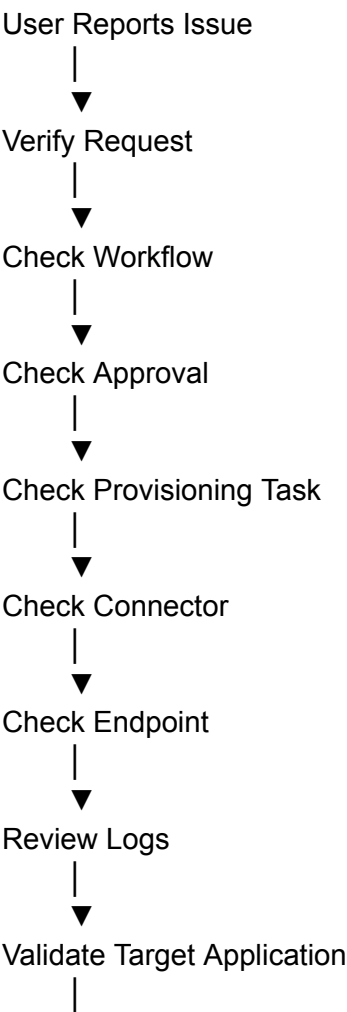
# Email Notification Issues

| Problem            | Cause             | Solution                  |
|--------------------|-------------------|---------------------------|
| Email not received | SMTP issue        | Verify SMTP               |
| Wrong recipient    | Incorrect mapping | Review email mapping      |
| Variables empty    | Template issue    | Correct dynamic variables |
| Delayed email      | Mail queue        | Review mail server        |

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## Complete Troubleshooting Workflow

Whenever an issue occurs, troubleshoot in this order:



▼  
Notify User

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## Real Project Example

**Company:** ABC Bank

Rahul requests Salesforce access.

### Issue

Manager approves.

No Salesforce account is created.

### Investigation

1. Workflow – Successful.
2. Approval – Completed.
3. Provisioning Task – Failed.
4. Connector Logs – Authentication Failed.
5. Service account password had expired.
6. Password updated.
7. Provisioning retried.
8. Salesforce account created successfully.

**Root Cause:** Expired connector service account password.

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## General Best Practices Checklist

- ✓ Use reusable workflows.
- ✓ Configure SoD policies.
- ✓ Keep request forms simple.
- ✓ Enable notifications.
- ✓ Test every workflow before production.

- ✓ Use Dynamic Peer Groups.
  - ✓ Monitor scheduled jobs.
  - ✓ Review logs regularly.
  - ✓ Enable audit trails.
  - ✓ Apply the Principle of Least Privilege.
  - ✓ Use JITAM for privileged access.
  - ✓ Keep identity data accurate.
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## Interview Questions

### **Q1. What are the best practices for configuring Access Requests?**

Use standard workflows, enable business justification, configure SoD checks, use start/end dates for temporary access, keep the access catalog clean, enable notifications, and monitor scheduled jobs.

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### **Q2. What should you check if an application is not visible in the Access Catalog?**

Verify that the application or entitlement is requestable, the Request Option is enabled, the Security System and Endpoint are active, and the application is correctly configured.

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### **Q3. How do you troubleshoot a workflow that does not start?**

Check whether the workflow is assigned, review approval rules, verify manager/resource owner mappings, and inspect workflow logs.

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### **Q4. Why might a Provisioning Task remain in the Pending state?**

Possible reasons include a stopped provisioning job, a connector issue, a queue backlog, or incomplete approvals.

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#### **Q5. Why are recommendations not displayed?**

The Recommendations Job may not have run, Intelligent Recommendations may be disabled, or Dynamic Peer Groups may not exist.

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#### **Q6. Why is the confidence score missing?**

Trust Modelling is disabled or the Trust Modelling Job has not been executed.

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#### **Q7. How do you troubleshoot email notification failures?**

Check SMTP configuration, credentials, firewall/network connectivity, email templates, recipient mappings, and notification logs.

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#### **Q8. What are common JITAM issues?**

Access not being automatically removed, incorrect access duration, request workflow failures, or roles not being configured as requestable.

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#### **Q9. What logs should you review during troubleshooting?**

Review workflow logs, provisioning logs, connector logs, job logs, notification logs, and application/API logs depending on the issue.

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#### **Q10. What is the recommended troubleshooting order?**

1. Verify the request.
2. Check the workflow.
3. Verify approvals.
4. Review the provisioning task.



5. Check the connector.
  6. Verify the endpoint.
  7. Review logs.
  8. Validate changes in the target application.
- 

## Quick Revision

- Follow **standard and reusable workflows**.
- Always require **business justification** for sensitive access.
- Use **Start Date** and **End Date** for temporary access.
- Enable **SoD checks** to prevent conflicting access.
- Keep the **Access Catalog** clean and organized.
- Configure **Email Notifications** for all important events.
- Use **Dynamic Peer Groups** for better recommendations.
- Monitor scheduled jobs such as **Recommendations, Trust Modelling, Provisioning, Import, and Reconciliation**.
- Troubleshoot in the order: **Request → Workflow → Approval → Provisioning Task → Connector → Endpoint → Logs → Target Application**.
- Use **JITAM** for privileged access instead of permanent administrator permissions.
- Regularly review logs and audit trails to quickly identify and resolve issues.